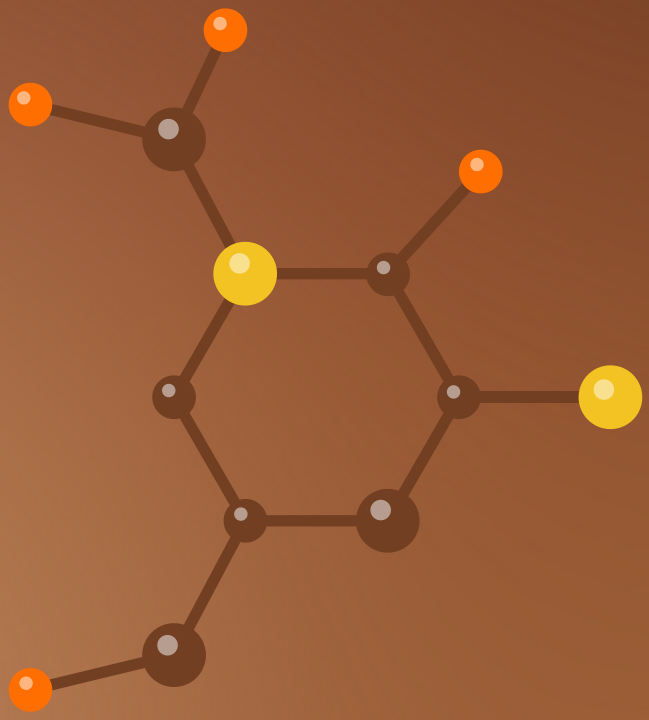
ALLEN CAREER INSTITUTE
path to success
KOTA (RAJASTHAN)



ORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Carbonyl Compound

English Medium

CARBONYL COMPOUND

EXERCISE # O-1

1. Arrange these compounds in decreasing order of reactivity for the nucleophilic attack :

(I) Acid chloride (II) Aldehyde (III) Ketone (IV) Ester

Select the correct answer from the codes given below :

(A) I > II > III > IV (B) IV > III > II > I (C) III > II > I > IV (D) I > IV > II > III

CL0001

2. In the given reaction

$$CH_3-CH_2-\overset{O}{\overset{\|}{C}}-CH_2COOC_2H_5 \xrightarrow{[X]} (A) \xrightarrow[(ii)\ H_2O/H^{\oplus}]{(i)\ LiAlH_4} CH_3-CH_2-\overset{O}{\overset{\|}{C}}-CH_2-CH_2OH+C_2H_5OH$$

[X] will be :

(A) HCHO (B) $\begin{matrix}CH_2-OH\\|\\CH_2-OH\end{matrix} + H^{\oplus}$ (C) $\begin{matrix}CH_2-OH\\|\\CH_2-OH\end{matrix} + \overset{\ominus}{O}H$ (D) HCN

CL0002

3. In the given reaction:

$$C_6H_5-\overset{O}{\overset{\|}{C}}-H \xrightarrow{NH_2OH/H^{\oplus}} [X]$$

[X] will be:

(A) Only syn oxime (B) Only anti oxime

(C) Mixture of syn and anti oxime (D) Secondary amide

CL0003

4. Acetophenone can be obtained by the distillation of :

(A) $(C_6H_5COO)_2Ca$ (B) $(CH_3COO)_2Ca$

(C) $(C_6H_5COO)_2Ca$ and $(CH_3COO)_2Ca$ (D) $(C_6H_5COO)_2Ca$ and $(HCOO)_2Ca$

CL0004

5. Gem dihalide on hydrolysis gives :

(A) Vic diol (B) Gem diol

(C) Carbonyl compound (D) Carboxylic acid

CL0005

6. Acetal or ketal is:

(A) Vic dialkoxy compound (B) α, ω-dialkoxy compound

(C) α-alkoxy alcohol (D) Gem dialkoxy compound

CL0006

7. Cross cannizzaro reaction is example of :

(A) Redox reaction (B) Disproportionation

(C) Both (A) and (B) (D) Only oxidation

CL0007

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.&Eth.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket.(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65"

8. Acetaldehyde can be converted into $HOCH_2-\underset{\underset{CH_2OH}{|}}{\overset{\overset{CH_2OH}{|}}{C}}-CH_2OH$ by which reagent ?

(A) KOH
(B) KOH followed by LAH
(C) excess of HCHO and KOH
(D) KCN followed by SBH

CL0008

9. Which one of the combinations will give propanaldehyde on dry distillation ?

(A) $(C_6H_5COO)_2Ca$ and $(HCOO)_2Ca$
(B) $(CH_3COO)_2Ca$ and $(CH_3CH_2-COO)_2Ca$
(C) $(CH_3-CH_2-COO)_2Ca$ and $(HCOO)_2Ca$
(D) $(CH_3COO)_2Ca$ and $(CH_3COO)_2Ca$

CL0009

10. In the given reaction :

$$CH_3-\overset{\overset{O}{\|}}{C}-CH_3 \xrightarrow[\text{Distillation}]{\text{Conc.}H_2SO_4} [X]$$

[X] will be :

(A) Methyl oxide
(B) Phorone
(C) 1, 3, 5-Trimethylbenzene
(D) 2-Butyne

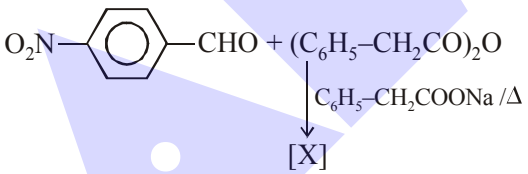
CL0010

11. Grignard reagents can not give carbonyl compounds with :

(A) CO_2
(B) RCOCl
(C) RCN
(D) RCOOR

CL0011

12. The product of the reaction :



will be :

(A) $C_6H_5-CH=CH-COOH$
(B) $NO_2-C_6H_4-CH=CH-COOH$
(C) $C_6H_5-CH=\underset{\underset{CH_3}{|}}{C}-COOH$
(D) $NO_2-C_6H_4-CH=\underset{\underset{C_6H_5}{|}}{C}-COOH$

CL0012

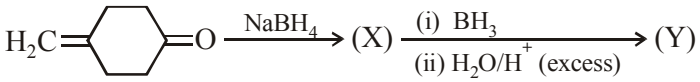
13. Cyanohydrin of which compound on hydrolysis will give lactic acid ?

(A) C_6H_5CHO
(B) HCHO
(C) CH_3CHO
(D) CH_3-CH_2-CHO

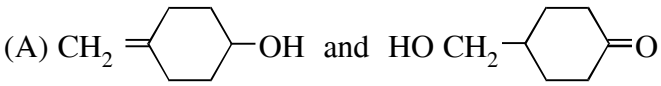
CL0013

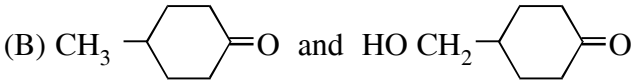
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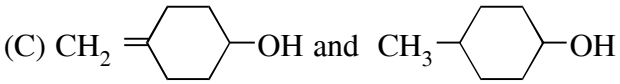
14. In the given reaction :

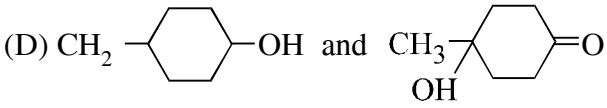


(X) and (Y) are :

(A) 

(B) 

(C) 

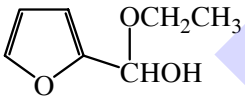
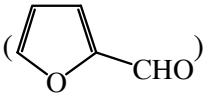
(D) 

CL0014

15. Acetaldehyde cannot give :

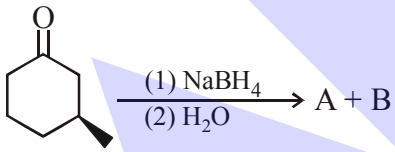
(A) Iodoform test (B) Lucas test (C) Benedict test (D) Tollens test

CL0015

16. Compound  formed by the reaction of furfural () with ethanol is :

(A) an aldol (B) an acetal (C) a ketal (D) a hemiacetal

CL0016

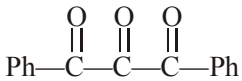
17. 

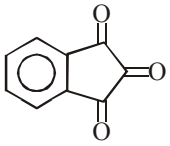
Identify relationship between A & B products ?

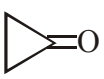
(A) Diastereoisomers (B) Enantiomers (C) Positional isomer (D) Identical

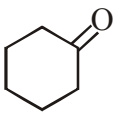
CL0017

18. Which of the following does not form a stable hydrate by the addition of H_2O?

(A) 

(B) 

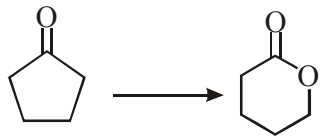
(C) 

(D) 

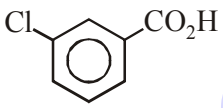
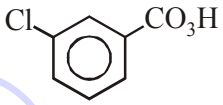
CL0018

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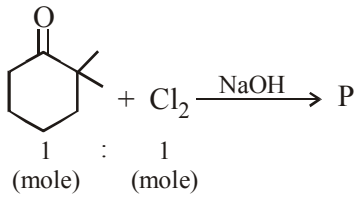
19. The conversion



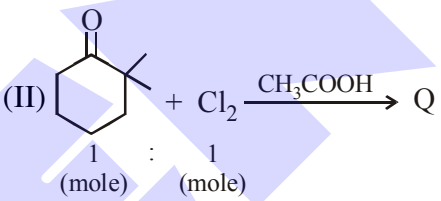
can be effected by using the reagent

(A) Tollen's reagent (B) O_3 (C)  (D) 

CL0019

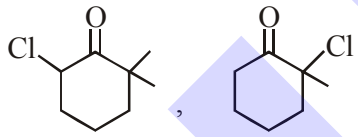
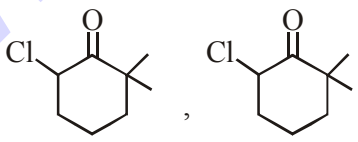
20. (I) 

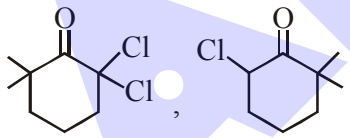
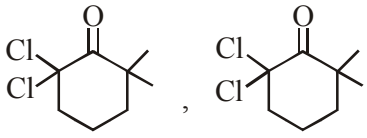
1 (mole) : 1 (mole)

(II) 

1 (mole) : 1 (mole)

Organic product P & Q are respectively -

(A)  (B) 

(C)  (D) 

CL0020

21. Total number of stereoisomers of major product (Q) are :

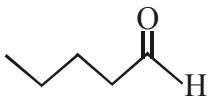
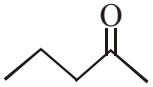
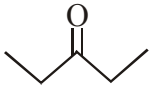
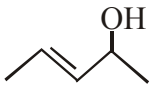


(A) 0 (B) 4 (C) 8 (D) 2

CL0021

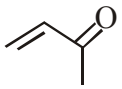
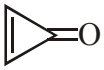
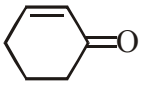
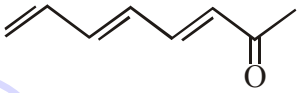
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22. An organic compound (A), $C_5H_{10}O$, reacts with hydrazine to form a hydrazone derivative (B). The hydrazone (B) on being heated with KOH at about 180ºC, gives n-pentane. The compound (A) does not respond positively to Tollen's reagent and to the iodoform test. The compound (A) is

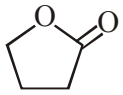
(A)  (B)  (C)  (D) 

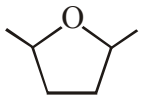
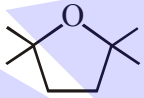
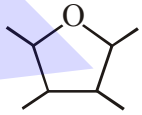
CL0022

23. The compound having the highest dipole moment is :

(A)  (B)  (C)  (D) 

CL0023

24.  $\xrightarrow[(2)\ H_2O]{(1)\ \text{Excess MeMgCl}}$ (A) $\xrightarrow[H_2SO_4]{\text{conc.}}$ 'B' Identify 'B' product ?

(A)  (B)  (C)  (D) 

CL0024

Question No. 25 to 27 (3 questions)

An alkene (A) $C_{16}H_{16}$ on ozonolysis gives only product (B) C_8H_8O. (B) also can be obtained by hydrolysis of the product obtained by reaction between cyano benzene and CH_3MgBr. (A) can show geometrical isomerism and it can decolourise Br_2 water. (B) on treatment with SeO_2 produces (C)

25. Which is not correct about (A) ?

(A) A is optically inactive

(B) On catalytic hydrogenation 'trans' form of A produces racemic mixture

(C) A can be prepared by Witting reaction on acetophenone with $Ph_3P = C(CH_3)Ph$.

(D) On treatment with per acid followed by hydrolysis 'trans' form of A produces racemic mixture

CL0025

26. Which is not correct about B ?

(A) It gives iodoform test

(B) On treatment with $LiAlH_4$, H_2O it produces a compound which also responds to iodoform test.

(C) It gives Tollen's test

(D) On treatment with NH_2NH_2 followed by alc. KOH at high temperature, it produces ethyl benzene

CL0026

27. Which is not correct about C?

(A) On treatment with $NaBH_4$ it will produce a diol.

(B) On treatment with OH^- (conc.) followed by acidification racemic mixture of a carboxylic acid is obtained

(C) It gives Tollen's test

(D) It can take part in aldol condensation

CL0027

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Hdb-Alc.&Et.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65"

(Question No. 28 & 29)

Questions given below consist of two statements each printed as Assertion (A) and Reason (R); while answering these questions you are required to choose any one of the following four responses:

(A) If both (A) and (R) are true and (R) is the correct explanation of (A)

(B) If both (A) and (R) are true but (R) is not correct explanation of (A)

(C) If (A) is true but (R) is false

(D) If (A) is false and (R) is true

28. **Assertion** : Benzaldehyde with HCN gives two isomeric compounds

Reason : Both nitrile and isonitrile compounds are possible when HCN reacts with carbonyl group.

CL0028

29. **Assertion :** $Cl_3C-\overset{\overset{O}{\|}}{C}-H \xrightarrow{NaOH} Cl_3C—CH_2OH + Cl_3C—COONa$

Reason : There are no α–H in this compound, so it can't give aldol.

CL0029

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdlo-Alc & Et - Hydr.-Carb.-Oxi.-Red.- Ald & Ket-(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65"

EXERCISE # O-2

1. Two isomeric ketones, 3-pentanone and 2-pentanone can be distinguished by :

(A) I_2 / NaOH (B)$NaSO_3H$ (C) NaCN / HCl (D) 2,4-DNP

CL0030

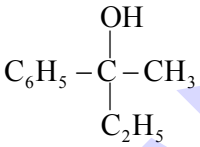
2. An optically inactive alcohol (A) $C_6H_{12}O$ is oxidized by MnO_2 to produce optically inactive carbonyl compound while reduction of (A) by H_2/Ni produces optically active compound. Possible structure(s) of alcohol is/are

(A) Hex-2-ene-1- ol (B) Hex-3-ene-2- ol

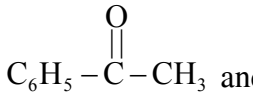
(C) 2-Methyl pent-2 - ene-1-ol (D) 3-Methyl pent-2 - ene-1-ol

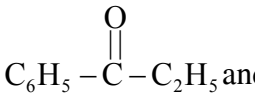
CL0031

3. Consider the structure of given alcohol:



This alcohol can be prepared from:

(A)  and C_2H_5MgBr (B)  and C_6H_5MgBr

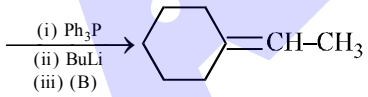
(C)  and CH_3MgBr (D)  and C_2H_5MgCl

CL0032

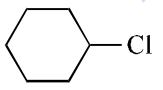
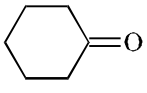
4. Which of the following compounds will not give aldol condensation :

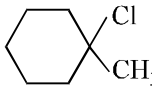
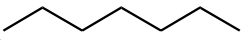
(A) Acetaldehyde (B) Formaldehyde (C) Pivaldehyde (D) Crotonaldehyde

CL0033

5. (A) 

In above reaction (A) and (B) will respectively be

(A)  & CH_3CHO (B) CH_3CH_2Cl & 

(C)  & HCHO (D)  & CH_3CHO

CL0034.

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.&Eth.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\04_Carbonyl\101_Theory+Ex.p65"

6. Stability of hydrates of carbonyl compounds depends on:

(A) Steric hindrance (B) Presence of –I group on gemdiol carbon

(C) Intramolecular hydrogen bonding (D) angle strain in carbonyl compound

CL0035

7. Which of the following can be used for protection of carbonyl group

(A) $CH_2OH–CH_2OH$ / $H^{\oplus}$ (B) $CH_2OH–CH_2–CH_2OH$ / $H^{\oplus}$

(C) $HS–(CH_2)_3–SH$ (D) $CH_2OH–CH_2–CHO$

CL0036

8. Which of the following(s) will form stable hemiketal :

(A) $Ph-\overset{\overset{O}{\|}}{C}-Ph$ (B) $HO-(CH_2)_3-\overset{\overset{O}{\|}}{C}-CH_3$

(C) $CH_2OH-\overset{\overset{O}{\|}}{C}-(CHOH)_3-CH_2-OH$ (D) $H_3C-O–CH_2–CH_2–CH_2–\overset{\overset{O}{\|}}{C}–CH_3$

CL0037

9. Mixture of Ph–CHO & HCHO is treated with NaOH then Cannizzaro reaction involves:

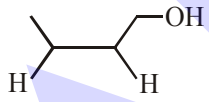
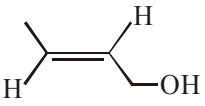
(A) Oxidation of HCHO (B) Reduction of HCHO

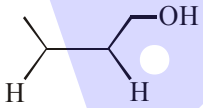
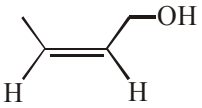
(C) Oxidation of Ph–CHO (D) Reduction of Ph–CHO

CL0038

10. Final product in the given reaction sequence is :

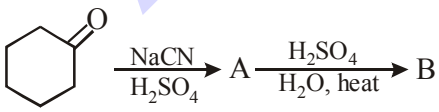
$$CH_3–C\equiv CH \xrightarrow{PhMgBr} \bigcirc + [A] \xrightarrow[ii)\ H^{\oplus}]{i)\ H-\overset{\overset{O}{\|}}{C}-H} [B] \xrightarrow[Pd/BaSO_4]{H_2} [C]$$

(A)  (B) 

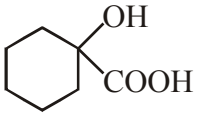
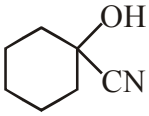
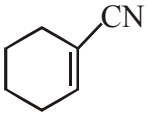
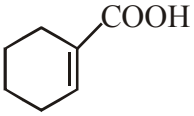
(C)  (D) 

CL0039

11. Consider the following sequence of reactions.



The major product (B) is :

(A)  (B)  (C)  (D) 

CL0040

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12. In the reaction

$$(CH_3)_2CHNO_2 + HCHO \xrightarrow{NaOH}$$

the major product is

(A) $(CH_3)_2CHCH_2ONO_2$

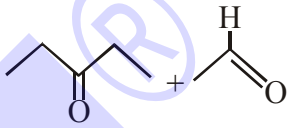
(B) $(CH_3)_2C(NO_2)(CH_2OH)$

(C) $(CH_3)_2CH-CH(O)CHNO_2$

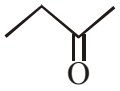
(D) $(CH_3)_2CH-CH(O)-CH_2NO_2$

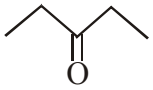
CL0041

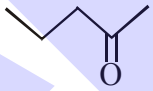
13. Consider the following sequence of reactions.

Ketone A $\xrightarrow[2.\ H_2O]{1.\ C_2H_5MgBr}$ B $\xrightarrow[-H_2O]{H_2SO_4,\ heat}$ C $\xrightarrow[2.\ Zn,\ H_2O]{1.\ O_3}$ 

The ketone (A) is :

(A) 

(B) 

(C) 

(D) 

CL0042

14. Which of the following reactions will give(s) 2° alcohol as a major product :

(A) $CH_3-CH_2-C(=O)-NH_2 \xrightarrow[(ii)\ NaNO_2/HCl]{(i)\ LAH}$

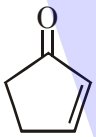
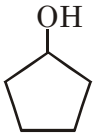
(B) $H-C(=O)-OR \xrightarrow[(ii)\ H^+]{(i)\ CH_3MgX(excess)}$

(C) $H_3C-HC(O)CH_2 \xrightarrow[(ii)\ H^+]{(i)\ RMgX}$

(D) $CH_3-C(=O)-Cl \xrightarrow[H^+]{CH_3MgX\ (excess)}$

CL0043

15. Match list-I with list-II :

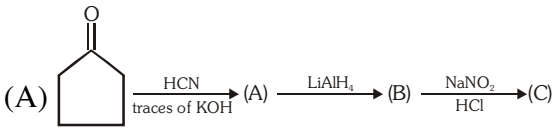
List - I	**List - II**
(A)  $\xrightarrow{NaBH_4}$	(P) Acetal
(B) $C_6H_5CHO + Ph-NH_2 \xrightarrow{H^\oplus}$	(Q) Schiff's base
(C) $C_6H_5COCH_3 + CH_3-CH_2-NH_2 \xrightarrow{H^\oplus}$	(R) 
(D) $RCHO + 2RCH_2OH \xrightarrow{H^\oplus}$	(S) Imine

CL0044

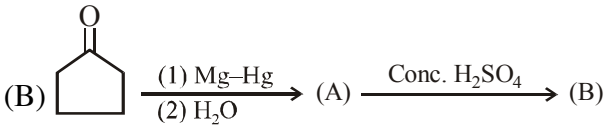
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.&Eth.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket.(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65"

EXERCISE # S-1

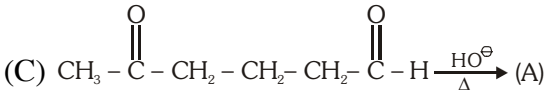
1. **Column - I** **Column - II**

(A) 

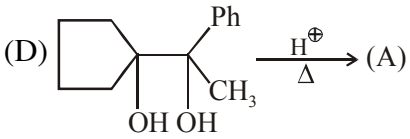
(P) Formation of six member ring takes place

(B) 

(Q) Final product is Ketone

(C) 

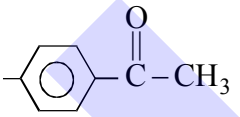
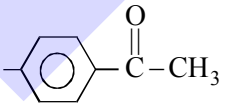
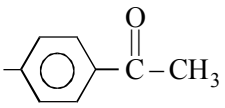
(R) Final product formed will give positive Idoform test

(D) 

(S) Final product formed will react with 2,4-DNP. (2,4-Di-nitrophenyl hydrazine)

CL0045

2. Arrange the following compounds in decreasing order of K_e for hydrate formation.

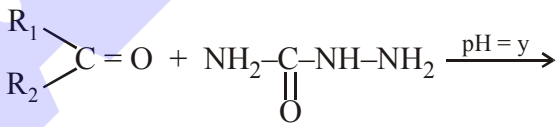
(1) $C_6H_5COCH_3$ (2)  (3)  (4) 

CL0046

Paragraph for 03 to 04

Two reactions which are example of nucleophilic attack are given as below .

Reaction - I : 

Reaction - II : 

3. Value of x is :

(A) $x \leq 4.5$ (B) $x = 6$ (C) $x > 7$ (D) Can't decide

CL0047

4. Value of y is :

(A) $x = 4.5$ (B) $x = 1.5$ (C) $x = 7$ (D) $x = 9$

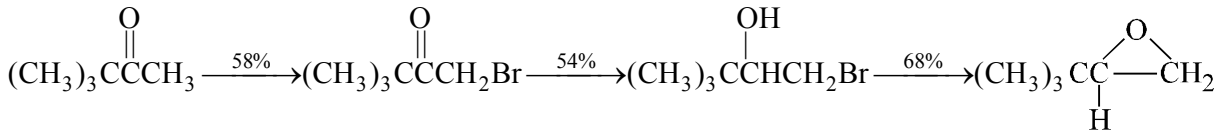
CL0047

5. Some Grignard reagents react with ethyl orthoformate, followed by acidic hydrolysis, to give aldehydes. Propose mechanisms for the two steps in this synthesis.

CL0048

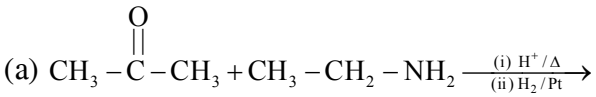
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hlkc Alc & Et - Hydr - Carb - Oxi - Red - Alld & Ket-(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65"

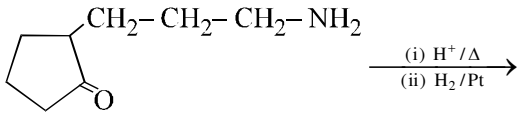
6. A synthesis that begins with 3,3-dimethyl-2-butanone gives the epoxide shown. Suggest reagents appropriate for each step in the synthesis.

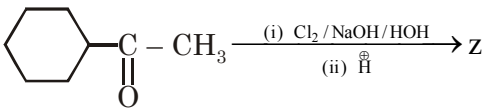


CL0049

7. Predict the organic products :

(a) 

(b) 

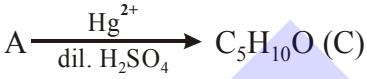
(c) 

CL0050

Paragraph for No.8 to 9

A(Hydrocarbon) (C ⇒ 88.24%) [Molecular weight of A = 68]



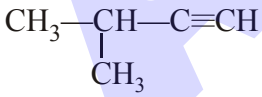


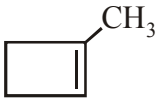


8. 'A' can be :

(A) $H_3C–H_2C–C\equiv C–CH_3$

(B) $CH_3–CH_2–CH_2–C\equiv C–H$

(C) 

(D) 

CL0051

9. Correct statement reagarding C is.

(A) C reacts with fehling solution to give red ppt.

(B) C gives +ve iodoform test

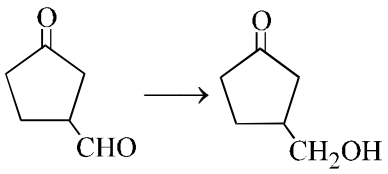
(C) C give –ve 2, 4, D.N.P test

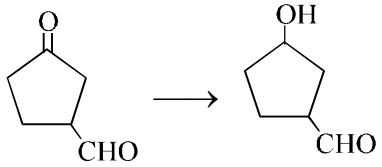
(D) C is aldehyde

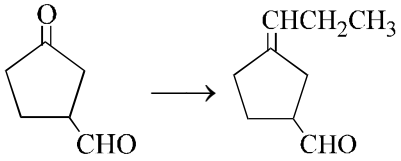
CL0051

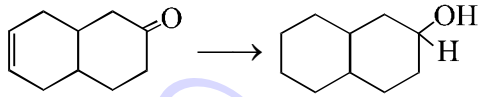
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Hdb-Alc.&Eth.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65"

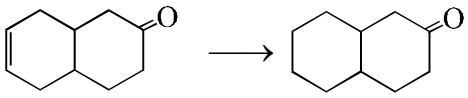
10. Show how you would accomplish the following syntheses efficiently and in good yield. You may use any necessary reagents.

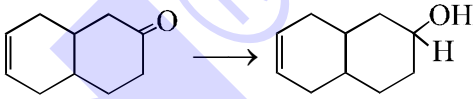
(a) 

(b) 

(c) 

(d) 

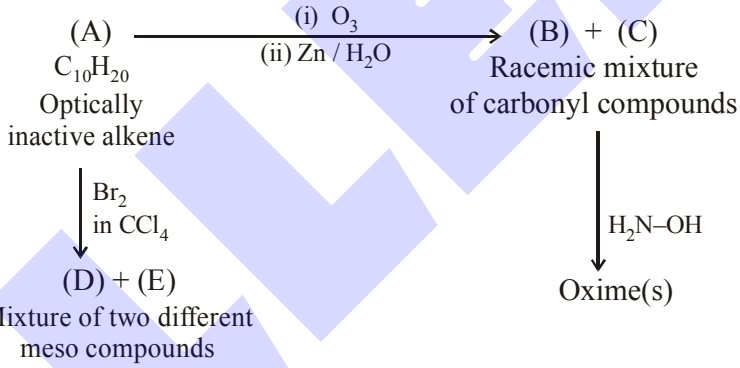
(e) 

(f) 

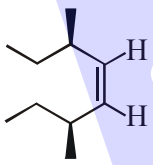
CL0052

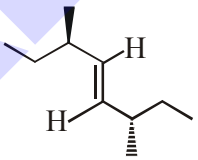
Paragraph for No. 11 to 12

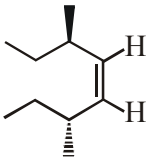
In given reaction sequence

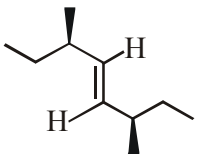


11. Alkene A is :

(A) 

(B) 

(C) 

(D) 

CL0053

12. How many total oxime(s) is/are obtained

(A) 1 (B) 2 (C) 3 (D) 4

CL0053

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdb-Alc & Et - Hydr.-Carb.-Oxi.-Red.-Ald & Ket-(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65

EXERCISE # MAINS

1. When $CH_2=CH-COOH$ is reduced with $LiAlH_4$, the compound obtained will be -
(1) $CH_3-CH_2-CH_2OH$ (2) CH_3-CH_2-CHO **[AIEEE-2003]**
(3) CH_3-CH_2-COOH (4) $CH_2=CH-CH_2OH$

CL0054

2. Which one of the following undergoes reaction with 50% sodium hydroxide solution to give the corresponding alcohol and acid ? **[AIEEE-2004]**
(1) Phenol (2) Benzaldehyde (3) Butanal (4) Benzoic acid

CL0055

3. Which one of the following is reduced with Zn-Hg/HCl to give the corresponding hydrocarbon **[AIEEE-2004]**
(1) Butan-2-one (2) Acetic acid (3) Acetamide (4) Ethyl acetate

CL0056

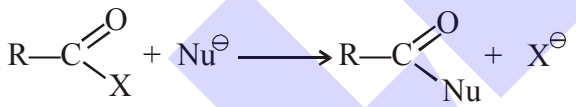
4. On mixing ethyl acetate with aqueous sodium chloride, the composition of the resultant solution is
(1) $CH_3COOC_2H_5 + NaCl$ (2) $CH_3COONa + C_2H_5OH$ **[AIEEE-2004]**
(3) $CH_3COCl + C_2H_5OH + NaOH$ (4) $CH_3Cl + C_2H_5COONa$

CL0057

5. The best reagent to convert pent–3–en–2–ol into pent –3–en –2–one is - **[AIEEE-2005]**
(1) Acidic dichromate (2) Acidic permanganate
(3) Pyridinium chloro-chromate (4) Chromic anhydride in glacial acetic acid

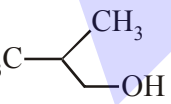
CL0058

6. Rate of the reaction- **[AIEEE-2005]**



is fastest when X is -
(1) NH_2 (2) Cl (3) OCOR (4) OC_2H_5

CL0059

7. Among the following the one that gives positive iodoform test upon reaction with I_2 and NaOH is-
(1) $CH_3CH_2CH(OH)CH_2CH_3$ (2) $C_6H_5CH_2CH_2OH$ **[AIEEE-2006]**
(3)  (4) $PhCHOHCH_3$

CL0060

8. In the following sequence of reactions
$CH_3CH_2OH \xrightarrow{P+I_2} A \xrightarrow[Ether]{Mg} B \xrightarrow{HCHO} C \xrightarrow{H_2O} D$, then compound 'D' is - **[AIEEE-2007]**
(1) Butanal (2) n–Butyl alcohol
(3) n–Propyl alcohol (4) Propanal

CL0061

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9. In the following sequence of reactions, the alkene affords the compound 'B' :- **[AIEEE-2008]**

$$CH_3CH = CHCH_3 \xrightarrow{O_3} A \xrightarrow[Zn]{H_2O} B.$$

The compound B is

(1) CH_3CH_2CHO (2) CH_3COCH_3 (3) $CH_3CH_2COCH_3$ (4) CH_3CHO

CL0062

10. Which of the following on heating with aqueous KOH, produces acetaldehyde ? **[AIEEE-2009]**

(1) CH_2ClCH_2Cl (2) CH_3CHCl_2 (3) CH_3COCl (4) CH_3CH_2Cl

CL0064

11. In Cannizzaro reaction given below :- **[AIEEE-2009]**

$$2PhCHO \xrightarrow{:\overset{\ominus}{O}H} PhCH_2OH + PhCO_2^{\ominus}$$

the slowest step is :-

(1) The abstraction of proton from the carboxylic group

(2) The deprotonation of $PhCH_2OH$

(3) The attack of $:\overset{\ominus}{O}H$ at the carboxyl group

(4) The transfer of hydride to the carbonyl group

CL0065

12. One mole of a symmetrical alkene on ozonolysis gives two moles of an aldehyde having a molecular mass of 44 u. The alkene is :- **[AIEEE-2010]**

(1) Ethene (2) Propene (3) 1-Butene (4) 2-Butene

CL0066

13. Ozonolysis of an organic compound 'A' produces acetone and propionaldehyde in equimolar mixture. Identify 'A' from the following compounds : - **[AIEEE-2011]**

(1) 2-Methyl - 1- pentene (2) 1-Pentene

(3) 2-Pentene (4) 2-Methyl-2-pentene

CL0068

14. Trichloroacetaldehyde was subjected to assumed Cannizzaro's reaction by using NaOH. The mixture of the products contains sodium trichloroacetate and another compound. The other compound is :- **[AIEEE-2011]**

(1) 2,2,2–Trichloropropanol (2) Chloroform

(3) 2,2,2–Trichloroethanol (4) Trichloromethanol

CL0069

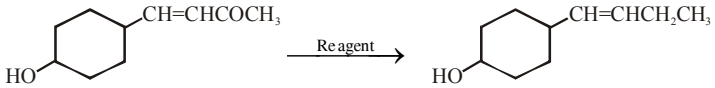
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdlo-Alc & Et - Hydr.-Carb.-Oxi.-Red.- Ald & Ket-(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65

15. Silver Mirror test is given by which one of the following compounds? **[AIEEE-2011]**

(1) Formaldehyde (2) Benzophenone (3) Acetaldehyde (4) Acetone

CL0070

16. In the given transformation, which of the following is the most appropriate reagent ? **[AIEEE-2012]**



(1) $NaBH_4$ (2) $NH_2NH_2, \overset{\ominus}{O}H$ (3) Zn – Hg / HCl (4) Na, $LiNH_3$

CL0071

17. Iodoform can be prepared from all except :- **[AIEEE-2012]**

(1) Isobutyl alcohol (2) Ethyl methyl ketone

(3) Isopropyl alcohol (4) 3-Methyl–2–butanone

CL0072

18. The major organic compound formed by the reaction of 1, 1, 1–trichloroethane with silver powder is :- **[JEE(Main)-2014]**

(1) 2-Butyne (2) 2-Butene (3) Acetylen (4) Ethene

CL0073

19. The most suitable reagent for the conversion of $R - CH_2 - OH \rightarrow R - CHO$ is :-

[JEE(Main)-2014]

(1) CrO_3 (2) PCC (Pyridinium chlorochromate)

(3) $KMNO_4$ (4) $K_2Cr_2O_7$

CL0074

20. In the reaction sequence **[JEE(Main)-2015]**

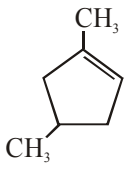
$2CH_3CHO \xrightarrow{OH^-} A \xrightarrow{\Delta} B$; the product B is:-

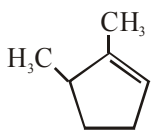
(1) $CH_3–CH=CH–CHO$ (2) $CH_3–CH_2–CH_2–CH_3$

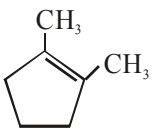
(3) $CH_3–CH_2–CH_2–CH_2–OH$ (4) $CH_3–\overset{\overset{O}{\|}}{C}–CH_3$

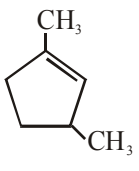
CL0076

21. Which compound would give 5-keto-2-methyl hexanal upon ozonlysis? **[JEE(Main) 2015]**

(1) 

(2) 

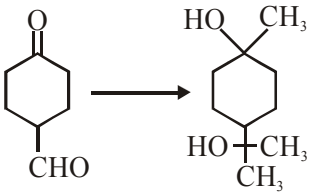
(3) 

(4) 

CL0077

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22. The correct sequence of reagents for the following conversion will be :- **[JEE(Main)-2017]**



(1) $[Ag(NH_3)_2]^+ OH^-$, H^+/CH_3OH, CH_3MgBr

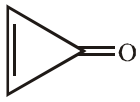
(2) CH_3MgBr, H^+/CH_3OH, $[Ag(NH_3)_2]^+ OH^-$

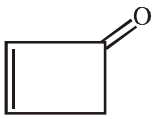
(3) CH_3MgBr, $[Ag(NH_3)_2]^+ OH^-$, H^+/CH_3OH

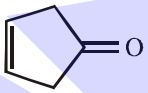
(4) $[Ag(NH_3)_2]^+ OH^-$, CH_3MgBr, H^+/CH_3OH

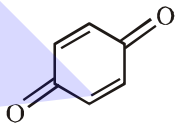
CL0078

23. Which of the following compounds will show highest dipole moment ? **[JEE(Main-on-line)-2017]**

(I) 

(II) 

(III) 

(IV) 

(1) (II) (2) (III) (3) (I) (4) (IV)

CL0079

24. Glucose on prolonged heating with HI gives : **[JEE(Main)-2018]**

(1) 1–Hexene (2) Hexanoic acid (3) 6-iodohexanal (4) n-Hexane

CL0080

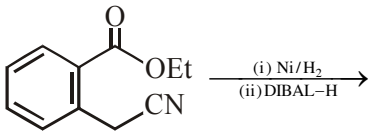
25. Which of the following compounds will most readily be dehydrated to give alkene under acidic condition? **[JEE(Main-on-line)-2018]**

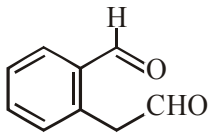
(1) 4-Hydroxypentan-2-one (2) 2-Hydroxycyclopentanone

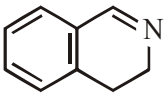
(3) 3-Hydroxypentan-2-one (4) 1-Pentanol

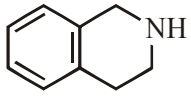
CL0081

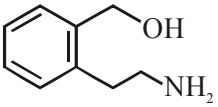
26. The major product of the following reaction is: **[JEE-Main(January)-2019]**



(1) 

(2) 

(3) 

(4) 

CL0082

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdlo-Alc & Et - Hydr.-Carb.-Oxi.-Red.- Ald & Ket-(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65

27. In the following reaction **[JEE-Main(January)-2019]**

$$\text{Aldehyde} + \text{Alcohol} \xrightarrow{HCl} \text{Acetal}$$

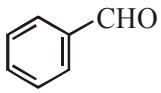
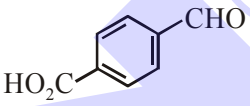
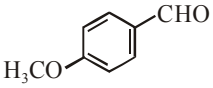
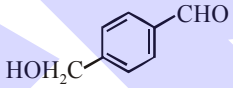
Aldehyde	Alcohol
HCHO	^{t}BuOH
CH_3CHO	MeOH

The best combinations is :

(1) HCHO and MeOH
(2) HCHO and ^{t}BuOH
(3) CH_3CHO and MeOH
(4) CH_3CHO and ^{t}BuOH

CL0083

28. The aldehydes which will not form Grignard product with one equivalent Grignard reagents are : **[JEE-Main(January)-2019]**

(A) 
(B) 
(C) 
(D) 

(1) (B), (C), (D)
(2) (B), (D)
(3) (B), (C)
(4) (C), (D)

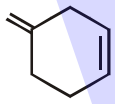
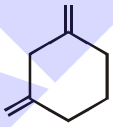
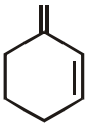
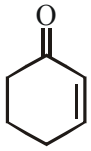
CL0084

29. An unsaturated hydrocarbon X absorbs two hydrogen molecules on catalytic hydrogenattion, and also gives following reaction : **[JEE-Main(Jan)-2020]**

$$X \xrightarrow[Zn/H_2O]{O_3} A \xrightarrow{[Ag(NH_3)_2]^+}$$

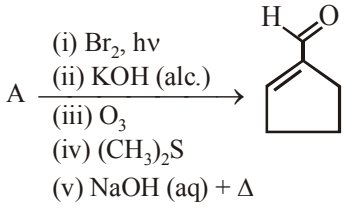
B(3 - oxo - hexanedicarboxylic acid)

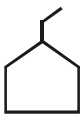
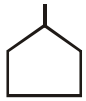
X will be :-

(1) 
(2) 
(3) 
(4) 

CL0085

30. In the following reaction A is : **[JEE-Main(Jan)-2020]**

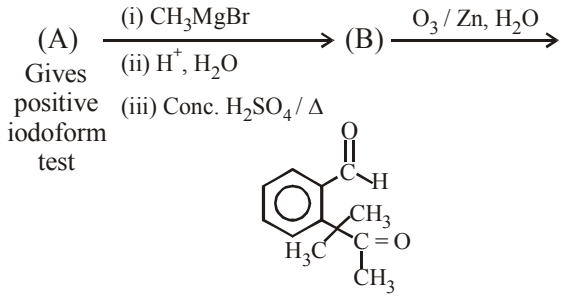


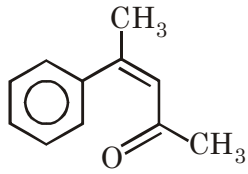
(1) 
(2) 
(3) 
(4) 

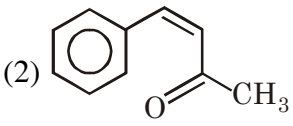
CL0086

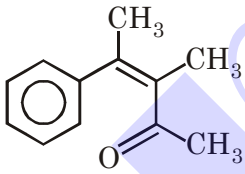
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.&Eth.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65"

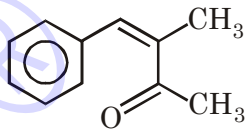
31. Identify (A) in the following reaction sequence : **[JEE-Main(Jan)-2020]**



(1) 

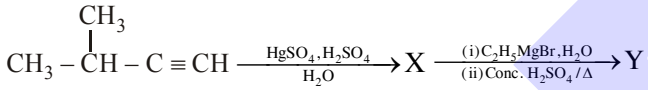
(2) 

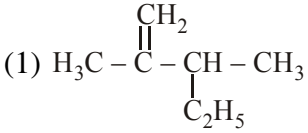
(3) 

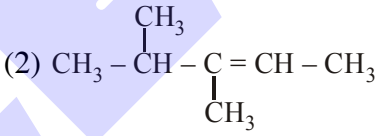
(4) 

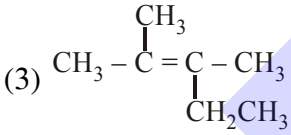
CL0088

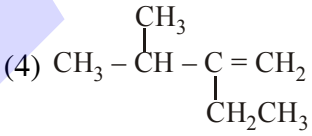
32. The major product (Y) in the following reactions is : **[JEE-Main(Jan)-2020]**



(1) 

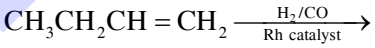
(2) 

(3) 

(4) 

CL0089

33. The major product of the following reaction is: **[JEE-Main-2021]**



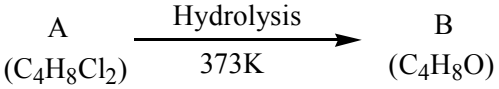
(1) $CH_3CH_2CH{=}CH{-}CHO$

(2) 

(3) $CH_3CH_2CH_2CH_2CHO$

(4) $CH_3CH_2CH_2CHO$

CL0116

34. 

[JEE-Main-2021]

B reacts with Hydroxyl amine but does not give Tollen's test. Identify A and B

(1) 1,1-Dichlorobutane and 2-Butanone

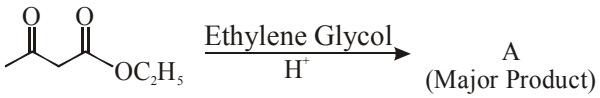
(2) 2,2-Dichlorobutane and Butanal

(3) 1,1-Dichlorobutane and Butanal

(4) 2,2-Dichlorobutane and 2-butan-one

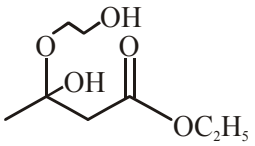
CL0117

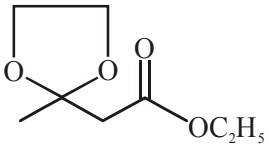
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hblo-Alc & Et - Hydr.-Carb.-Oxi.-Red.- Ald & Ket-(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65

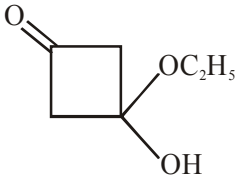
35. 

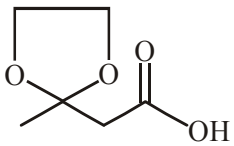
[JEE-Main-2021]

The product "A" in the above reaction is:

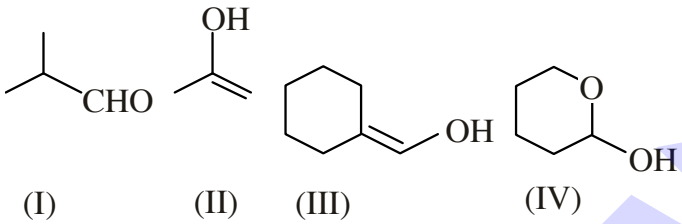
(1) 

(2) 

(3) 

(4) 

CL0118

36. 

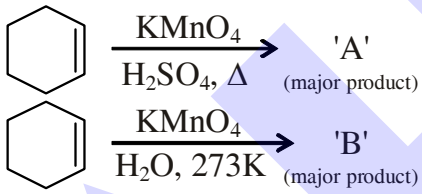
(I) (II) (III) (IV)

Which among the above compound/s does/do not form Silver mirror when treated with Tollen's reagent?

(1) (I), (III) and (IV) only (2) Only (IV) **[JEE-Main-2021]**

(3) Only (II) (4) (III) and (IV) only

CL0119

37. 

[JEE-Main-2021]

For above chemical reactions, identify the correct statement from the following:

(1) Both compound 'A' and compound 'B' are dicarboxylic acids

(2) Both compound 'A' and compound 'B' are diols

(3) Compound 'A' is diol and compound 'B' is dicarboxylic acid

(4) Compound 'A' is dicarboxylic acid and compound 'B' is diol

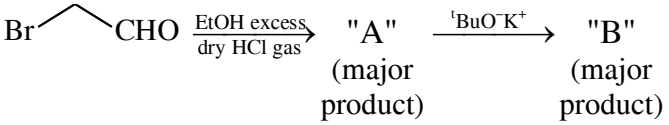
CL0120

38. A reaction of benzonitrile with one equivalent CH_3MgBr followed by hydrolysis produces a yellow liquid "P". The compound "P" will give positive_____. **[JEE-Main-2021]**

(1) Iodoform test (2) Schiff's test (3) Ninhydrin's test (4) Tollen's test

CL0121

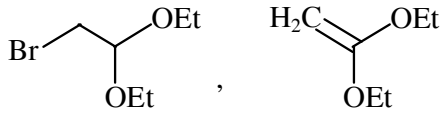
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Hdb-Alc.&Eth.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65"

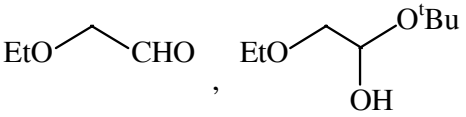
39. 

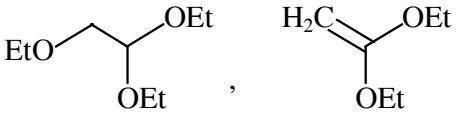
[JEE-Main-2021]

[where Et ⇒ $-C_2H_5$ ^tBu ⇒ $(CH_3)_3C-$]

Consider the above reaction sequence, Product "A" and Product "B" formed respectively are :

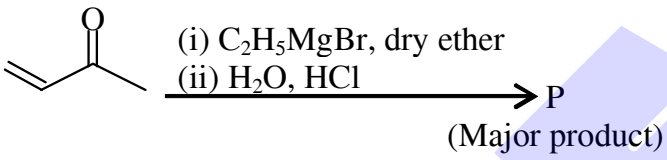
(1) 

(2) 

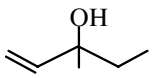
(3) 

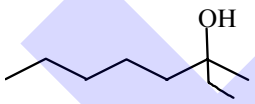
(4) 

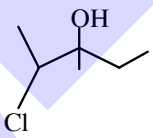
CL0122

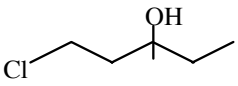
40. 

Consider the above reaction, the major product 'P' is: **[JEE-Main-2021]**

(1) 

(2) 

(3) 

(4) 

CL0123

41. An organic compound 'A' C_4H_8 on treatment with $KMnO_4/H^+$ yields compound 'B' C_3H_6O. Compound 'A' also yields compound 'B' an ozonolysis. Compound 'A' is :

(1) 2-Methylpropene

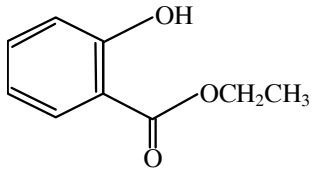
(2) 1-Methylcyclopropane **[JEE-Main-2021]**

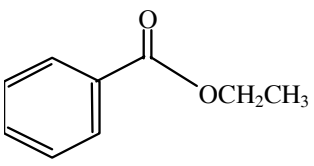
(3) But-2-ene

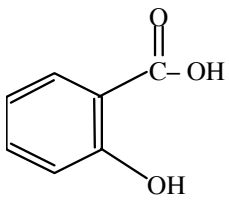
(4) Cyclobutane

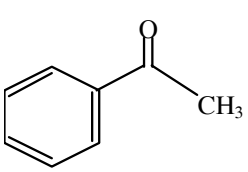
CL0124

42. Which one of the following compounds will give orange precipitate when treated with 2,4-dinitrophenyl hydrazine ? **[JEE-Main-2021]**

(1) 

(2) 

(3) 

(4) 

CL0125

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdb-Alc & Et - Hydr.-Carb.-Oxi.-Red.- Ald & Ket-(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65

43. Given below are two statements : **[JEE-Main-2021]**

Statement I : The nucleophilic addition of sodium hydrogen sulphite to an aldehyde or a ketone involves proton transfer to form a stable ion.

Statement II : The nucleophilic addition of hydrogen cyanide to an aldehyde or a ketone yields amine as final product.

In the light of the above statements, choose the most appropriate answer from the options given below :

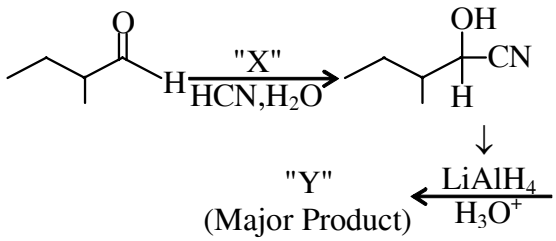
(1) Both Statement I and Statement II are true.

(2) Statement I is true but Statement II is false.

(3) Statement I is false but Statement II is true.

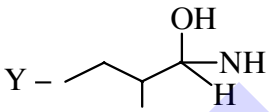
(4) Both Statement I and Statement II are false.

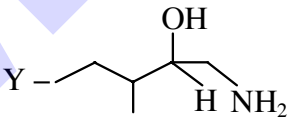
CL0126

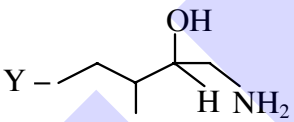
44. 

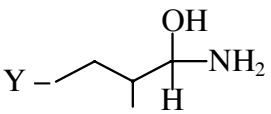
[JEE-Main-2021]

Consider the given reaction, Identify 'X' and 'Y' :

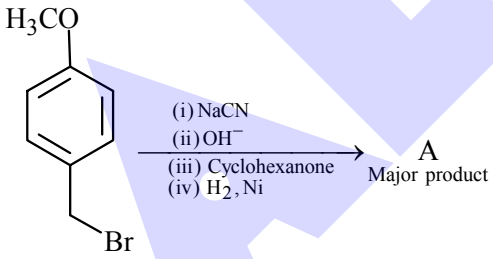
(1) X - $NaOH$ 

(2) X - HNO_3 

(3) X - $NaOH$ 

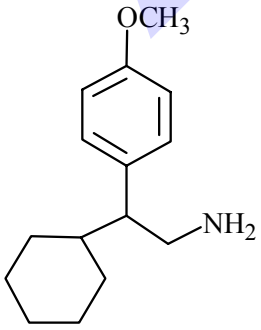
(4) X - HNO_3 

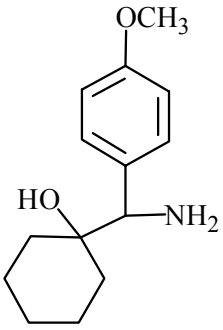
CL0127

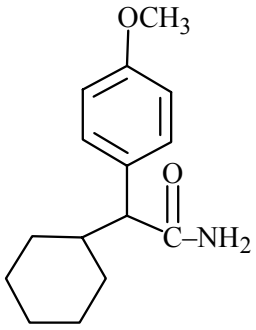
45. 

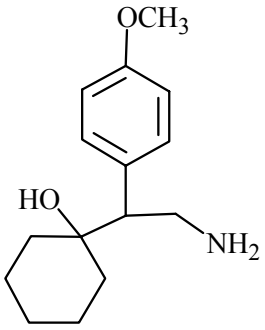
[JEE-Main-2022]

The major product of the above reaction is

(1) 

(2) 

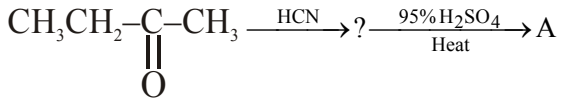
(3) 

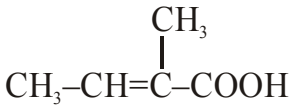
(4) 

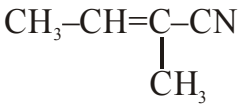
CL0130

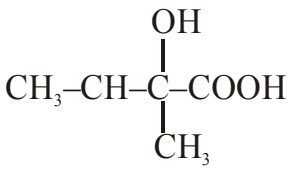
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.&Eth.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket.(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65"

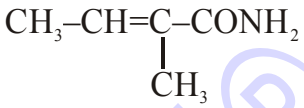
46. The final product 'A' in the following reaction sequence **[JEE-Main-2022]**



(1) 

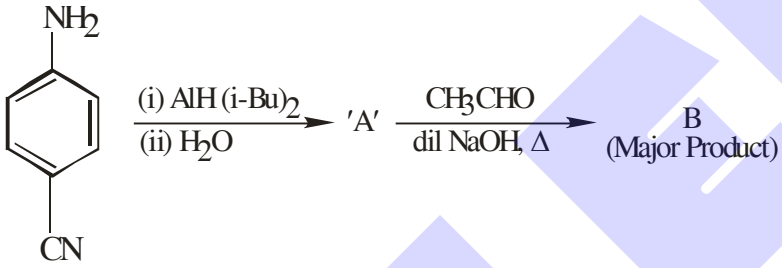
(2) 

(3) 

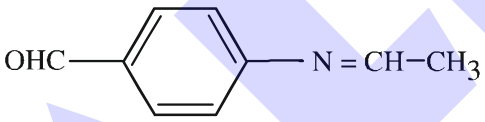
(4) 

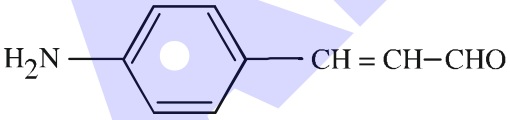
CL0131

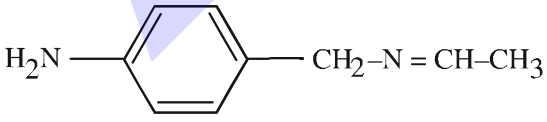
47. Consider the following reaction sequence : **[JEE-Main-2022]**

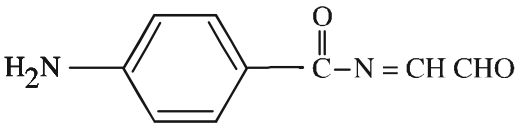


The product B is :

(1) 

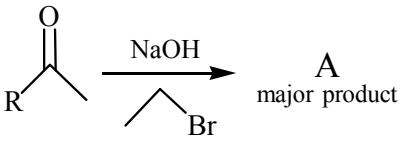
(2) 

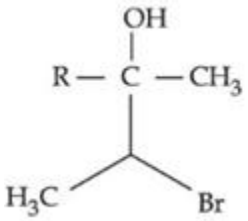
(3) 

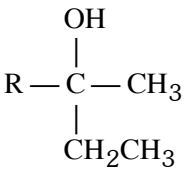
(4) 

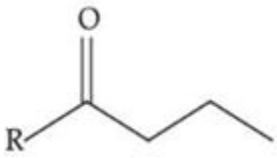
CL0132

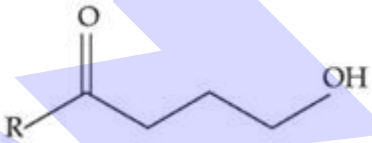
48. The structure of A in the given reaction is: **[JEE-Main-2022]**



(1) 

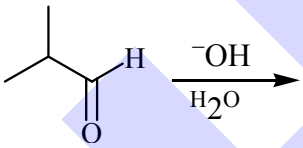
(2) 

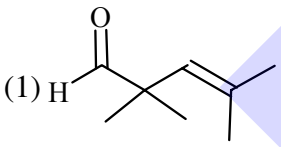
(3) 

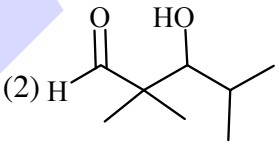
(4) 

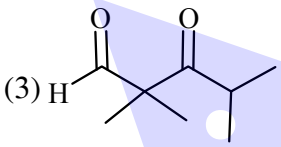
CL0133

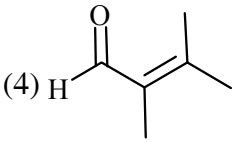
49. What is the major product of the following reaction? **[JEE-Main-2022]**



(1) 

(2) 

(3) 

(4) 

CL0134

50. Consider the following reactions **[JEE-Main(Jan)-2020]**

$$A \xrightarrow[(ii)H_3O^+]{(i)CH_3MgBr} B \xrightarrow[573K]{Cu} \text{2-methyl 2-butene}$$

The mass percentage of carbon in A is _____.

CL0087

51. In Tollen's test for aldehyde, the overall number of electron(s) transferred to the Tollen's reagent formula $[Ag(NH_3)_2]^+$ per aldehyde group to form silver mirror is_____.(Round off to the Nearest integer)

[JEE-Main-2021]

CL0128

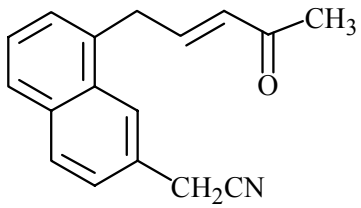
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#\Hdb-Alc.&Et.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket.-Hydr.-Carb.-Oxi.-Red.-(OC)\Eng\04_Carbonyl\101_Theory+Ex.p65"

52. The number of nitrogen atoms in a semicarbazone molecule of acetone is _____.

[JEE-Main-2021]

CL0129

53. Number of electrophilic centre in the given compound is ____ **[JEE-Main-2022]**



CL0135

54. The number of stereoisomers formed in a reaction of (±) Ph(C=O) C(OH)(CN)Ph with HCN is_____.

[JEE-Main-2022]

CL0136

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdlo-Alc & El - Hydr.-Carb.-Oxi.-Red.- Ald & Ket-(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65"

EXERCISE # IIT-QUESTIONS

1 Which of the following has the most acidic hydrogen: **[IIT 2000]**

(A) 3-hexanone

(B) 2,4-hexanedione

(C) 2,5-hexanedione

(D) 2,3-hexandione

CL0090

2 A mixture of benzaldehyde and formaldehyde on heating with aqueous NaOH solution gives:

(A) benzyl alcohol and sodium formate

(B) sodium benzoate and methyl alcohol

(C) sodium benzoate and sodium formate

(D) benzyl alcohol and methyl alcohol **[IIT 2001]**

CL0091

3 1-propanol & 2-propanol can be best distinguished by : **[IIT 2001]**

(A) Oxidation with alkaline $KMnO_4$ followed by reaction with Fehling solution

(B) Oxidation with acidic dichromate followed by reaction with Fehling solution

(C) Oxidation by heating with copper followed by reaction with Fehling solution

(D) Oxidation with concentrated H_2SO_4 followed by reaction with Fehling solution

CL0092

4 Compound A (molecular formula C_3H_8O) is treated with acidified potassium dichromate to form a product B (molecular formula C_3H_6O). B forms a shining silver mirror on warming with ammonical silver nitrate. B when treated with an aqueous solution of $H_2NCONHNH_2$. HCl and sodium acetate gives a product C. Identify the structure of C. **[IIT 2002]**

(A) $CH_3CH_2CH = NNHCONH_2$

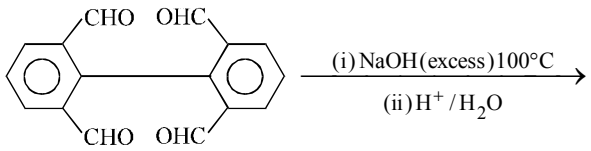
(B) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}=NNHCONH_2$

(C) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}=NCONHNH_2$

(D) $CH_3CH_2CH=NCONHNH_2$

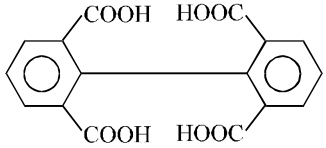
CL0093

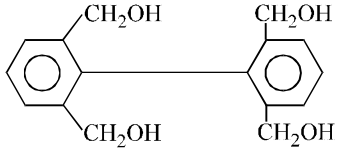
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Hdb-Alc & Et -Hydr -Carb -Oxi.-Red.- Ald & Ket-(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65"

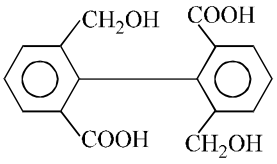
5 

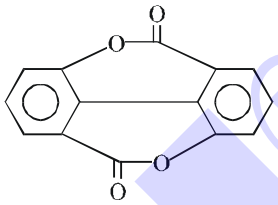
[IIT 2003]

any one of the products formed is :

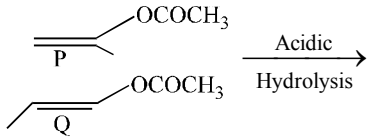
(A) 

(B) 

(C) 

(D) 

CL0094

6 

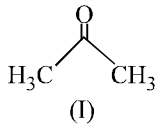
Products formed by P & Q can be differentiated by: **[IIT 2003]**

(A) 2, 4 DNP (B) Lucas reagent ($ZnCl_2$) conc. HCl

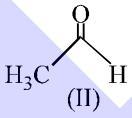
(C) $NaHSO_3$ (D) Fehlings solution

CL0095

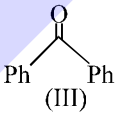
7 The order of reactivity of phenyl Magnesium Bromide with the following compounds is **[IIT 2004]**



(I)



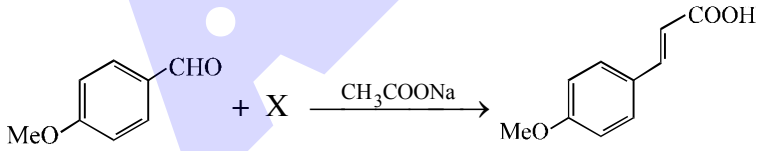
(II)



(III)

(A) II > III > I (B) I > III > II (C) II > I > III (D) All react with the same rate

CL0096

8 

[IIT 2005]

What is X?

(A) CH_3COOH (B) $BrCH_2$, COOH (C) $(CH_3CO)_2O$ (D) CHO–COOH

CL0097

9 The smallest ketone and its next homologue are reacted with NH_2OH to form oxime.

(A) Two different oximes are formed (B) Three different oximes are formed

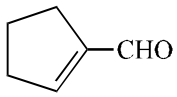
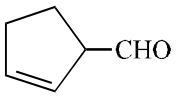
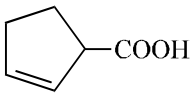
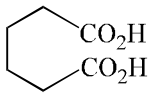
(C) Two oximes are optically active (D) All oximes are optically active **[IIT 2006]**

CL0098

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdlo-Alc & Et - Hydr - Carb - Oxi - Red - Ald & Ket - (OC)\Eng\04_Carbonyl\01_Theory+Ex.p65

10 Cyclohexene on ozonolysis followed by reaction with zinc dust and water gives compound E. Compound E on further treatment with aqueous KOH yields compound F. Compound F is

[IIT-JEE(ADV.)- 2007]

(A)  (B)  (C)  (D) 

CL0099

11 **Statement-1 :** Glucose gives a reddish-brown precipitate with Fehling's solution.
because
Statement-2 : Reaction of glucose with Fehling's solution gives CuO and gluconic acid.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement-2 is False.
(D) Statement-1 is False, Statement-2 is True. **[IIT-JEE(ADV.)- 2007]**

CL0100

12 Match the compounds/ion in column I with their properties/ reaction in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS.

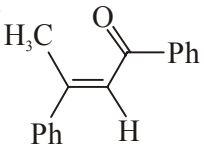
[IIT-JEE(ADV.)- 2007]

Column I	**Column II**
(A) C_6H_5CHO	(P) gives precipitate with 2,4-dinitrophenylhydrazine
(B) $CH_3C{\equiv}CH$	(Q) gives precipitate with $AgNO_3$
(C) CN^-	(R) is a nucleophile
(D) I^-	(S) is involved in cyanohydrin formation

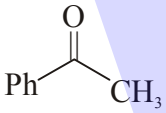
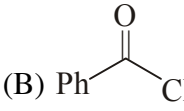
CL0101

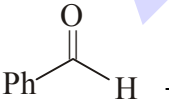
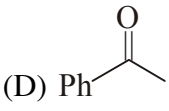
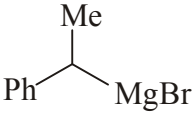
Paragraph for Question No. 13 to 15

A tertiary alcohol **H** upon acid catalysed dehydration gives a product **I**. Ozonolysis of **I** leads to compounds **J** and **K**. Compound **J** upon reaction with KOH gives benzyl alcohol and a compound **L**, whereas **K** on reaction with KOH gives only **M**.

M = 

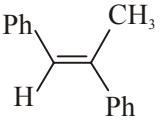
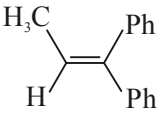
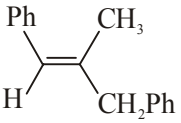
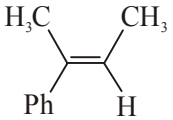
13 Compound **H** is formed by the reaction of **[IIT-JEE(ADV.)- 2008]**

(A)  + PhMgBr (B)  + $PhCH_2MgBr$

(C)  + $PhCH_2MgBr$ (D)  + 

CL0102

14 The structure of compound **I** is **[IIT-JEE(ADV.)- 2008]**

(A)  (B)  (C)  (D) 

CL0102

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Hdb-Alc.&Eth.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket.-Hydr.-Carb.-Oxi.-Red.-(OC)\Eng\04_Carbonyl\101_Theory+Ex.p65"

15 The structures of compounds **J**, **K** and **L**, respectively, are **[IIT-JEE(ADV.)- 2008]**

(A) $PhCOCH_3$, $PhCH_2COCH_3$ and $PhCH_2COO^-K^+$

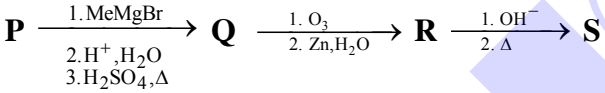
(B) PhCHO, $PhCH_2CHO$ and $PhCOO^-K^+$

(C) $PhCOCH_3$, $PhCH_2CHO$ and $CH_3COO^-K^+$

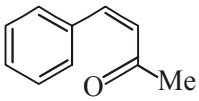
(D) PhCHO, $PhCOCH_3$ and $PhCOO^-K^+$ **CL0102**

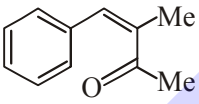
Paragraph for Question Nos. 16 to 38

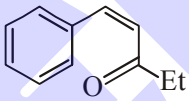
A carbonyl compound **P**, which gives positive iodoform test, undergoes reaction with MeMgBr followed by dehydration to give an olefin Ozonolysis of **Q** leads to a dicarbonyl compound **R**, which undergoes intramolecular aldol reaction to give predominantly **S**.

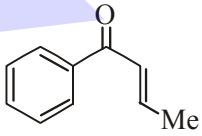


16 The structure of the carbonyl compound **P** is **[IIT-JEE(ADV.)- 2009]**

(A) 

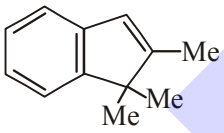
(B) 

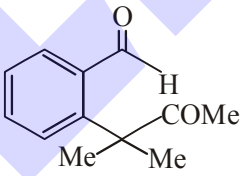
(C) 

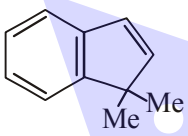
(D) 

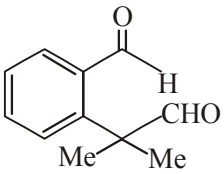
CL0103

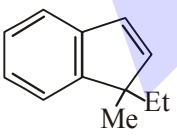
17 The structure of the products **Q** and **R**, respectively, are **[IIT-JEE(ADV.)- 2009]**

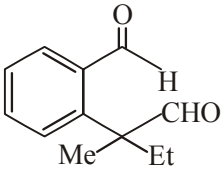
(A) 

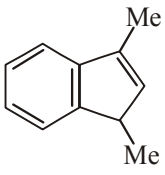
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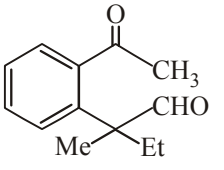
(B) 

,


(C) 

,


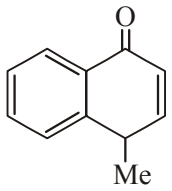
(D) 

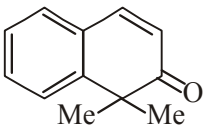
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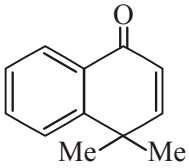
CL0103

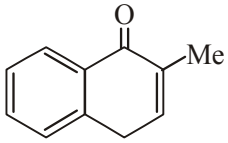
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hlde-Alc & Et - Hydr.-Carb.-Oxi.-Red.- Ald & Ket-(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65

18 The structure of the product **S** is **[IIT-JEE(ADV.)- 2009]**

(A) 

(B) 

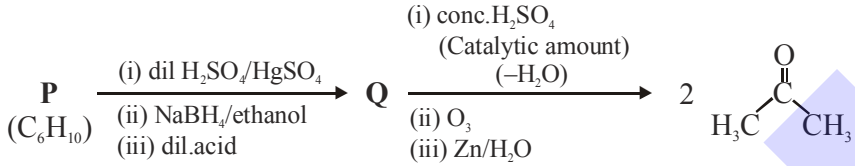
(C) 

(D) 

CL0103

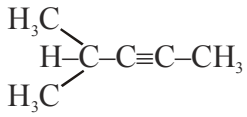
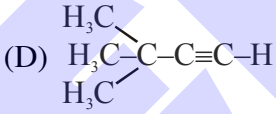
Paragraph for Questions Nos. 19 to 20

An acyclic hydrocarbon P, having molecular formula C_6H_{10}, gave acetone as the only organic product through the following sequence of reactions, in the which Q is an intermediate organic compound.



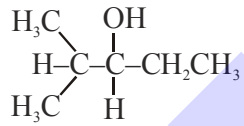
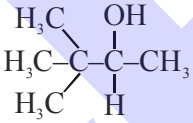
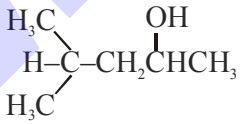
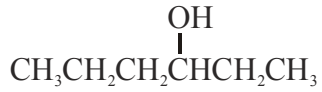
19 The structure of compound P is - **[IIT-JEE(ADV.)- 2011]**

(A) $CH_3CH_2CH_2CH_2–C\equiv C–H$ (B) $H_3CH_2C\equiv C–CH_2CH_3$

(C)  (D) 

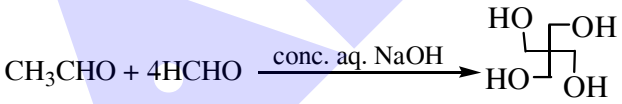
CL0104

20 The structure of the compound Q is - **[IIT-JEE(ADV.)- 2011]**

(A)  (B)  (C)  (D) 

CL0104

21 The number of aldol reaction(s) that occurs in the given transformation is **[IIT-JEE(ADV.)- 2012]**

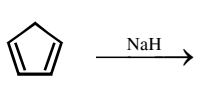


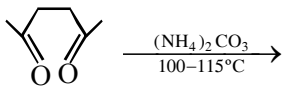
(A) 1 (B) 2 (C) 3 (D) 4

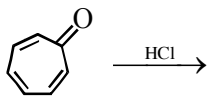
CL0105

22 Among P, Q, R and S, the aromatic compound(s) is / are : **[IIT-JEE(ADV.)- 2013]**







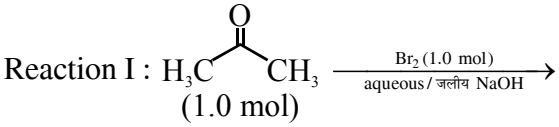


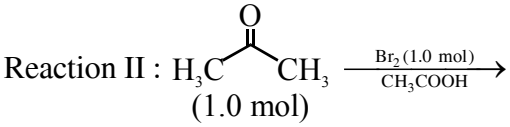
(A) P (B) Q (C) R (D) S

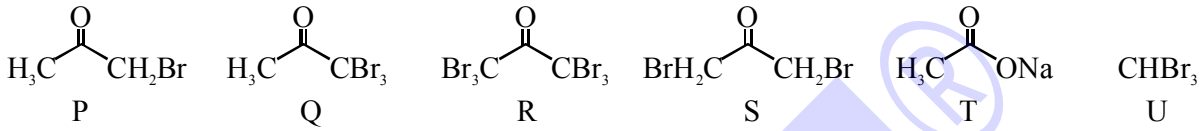
CL0106

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.&Et.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\04_Carbonyl\101_Theory+Ex.p65"

23 After completion of the reactions (I and II), the organic compound(s) in the reaction mixtures is(are) **[IIT-JEE(ADV.)- 2013]**

Reaction I : 

Reaction II : 



(A) Reaction I : P and Reaction II : P

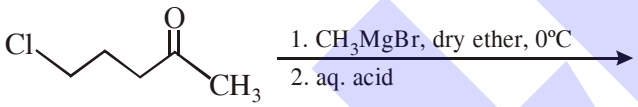
(B) Reaction I : U, acetone and Reaction II : Q acetone

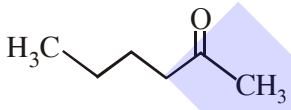
(C) Reaction I : T, U, acetone and Reaction II : P

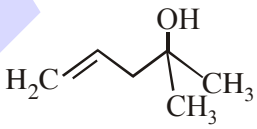
(D) Reaction I : R, acetone and Reaction II : S acetone

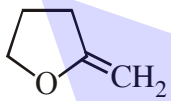
CL0107

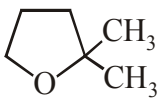
24 The major product in the following reaction is **[IIT-JEE(ADV.)- 2014]**



(A) 

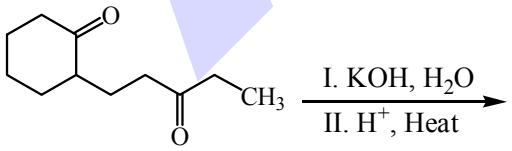
(B) 

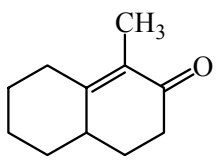
(C) 

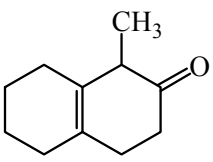
(D) 

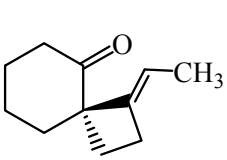
CL0108

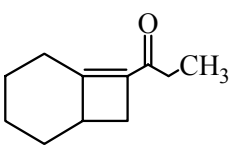
25 The major product of the following reaction is - **[IIT-JEE(ADV.)- 2015]**



(A) 

(B) 

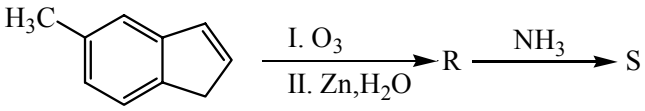
(C) 

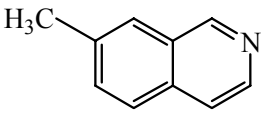
(D) 

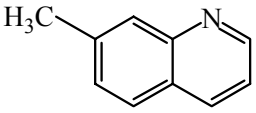
CL0109

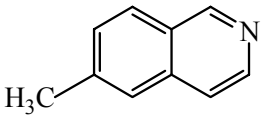
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdlc-Alc & El - Hydr.-Carb.-Oxi.-Red.- Ald & Ket-(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65

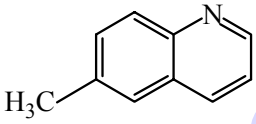
26 In the following reactions, the product S is - **[IIT-JEE(ADV.)- 2015]**



(A) 

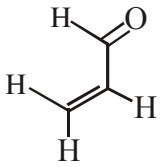
(B) 

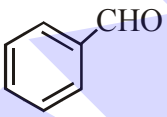
(C) 

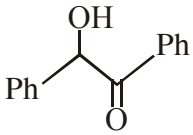
(D) 

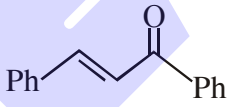
CL0110

27 Positive Tollen's test is observed for : **[IIT-JEE(ADV.)- 2016]**

(A) 

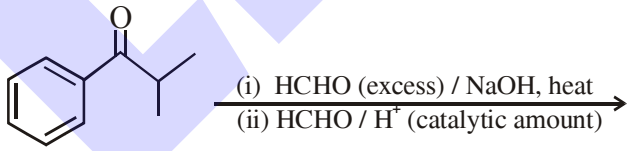
(B) 

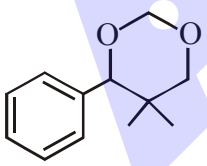
(C) 

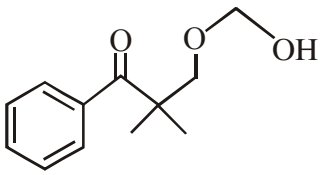
(D) 

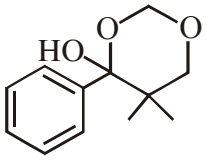
CL0111

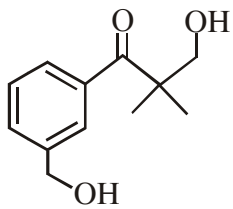
28 The major product of the following reaction sequence is : **[IITJEE(ADV.)-2016]**



(A) 

(B) 

(C) 

(D) 

CL0112

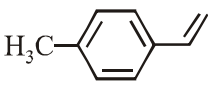
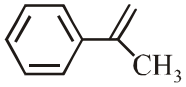
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.&Eth.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\04_Carbonyl\101_Theory+Ex.p65"

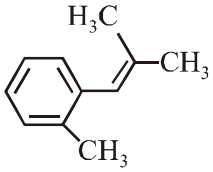
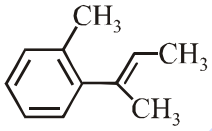
29 Compound **P** and **R** upon ozonolysis produce **Q** and **S**, respectively. The molecular formula of **Q** and **S** is C_8H_8O. **Q** undergoes Cannizzaro reaction but not haloform reaction, whereas **S** undergoes haloform reaction but not Cannizzaro reaction : **[IIT-JEE(ADV.)- 2017]**

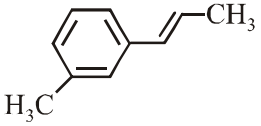
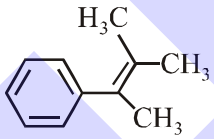
(i) P $\xrightarrow[\text{(ii) } Zn / H_2O]{\text{(i) } O_3 / CH_2Cl_2}$ Q (C_8H_8O)

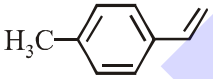
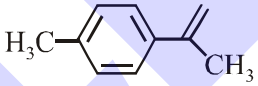
(ii) R $\xrightarrow[\text{(ii) } Zn / H_2O]{\text{(i) } O_3 / CH_2Cl_2}$ S (C_8H_8O)

The option(s) with suitable combination of P and R, respectively, is(are)

(A)  and 

(B)  and 

(C)  and 

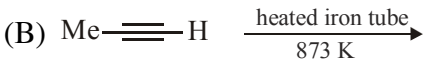
(D)  and 

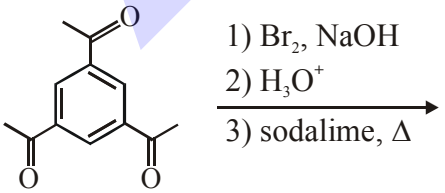
CL0113

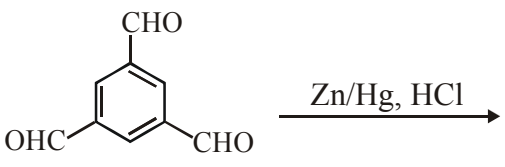
30. The reaction(s) leading to the formation of 1,3,5-trimethylbenzene is (are)

[IIT-JEE(ADV.)- 2018]

(A) 

(B) 

(C) 

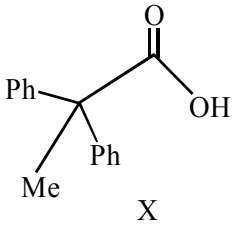
(D) 

CL0114

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdlo-Alc & Et - Hydr - Carb - Oxi - Red - Ald & Ket-(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65

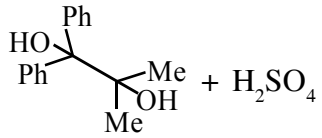
31. The desired product X can be prepared by reacting the major product of the reactions in LIST-I with one or more appropriate reagents in LIST-II. **[IIT-JEE(Adv.)-2018]**

(given, order of migratory aptitude: aryl > alkyl > hydrogen)

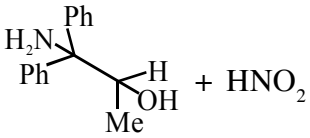


X

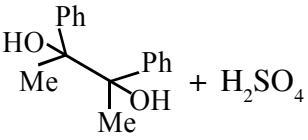
LIST-I **LIST-II**

P.  + H_2SO_4

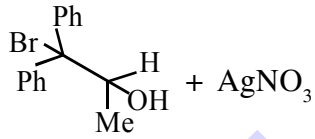
1. I_2, NaOH

 + HNO_2

2. $[Ag(NH_3)_2]OH$

R.  + H_2SO_4

3. Fehling solution

S.  + $AgNO_3$

4. HCHO, NaOH

5. NaOBr

The correct option is

(A) P → 1; Q → 2,3; R → 1,4; S → 2,4

(B) P → 1,5; Q → 3,4; R → 4,5; S → 3

(C) P → 1,5; Q → 3,4; R → 5; S → 2,4

(D) P → 1,5; Q → 2,3; R → 1,5; S → 2,3

CL0115

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Hdb-Alc.&Et.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\04_Carbonyl\101_Theory+Ex.p65"

ANSWER-KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	C	C	C	D	A	C	C	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	D	C	C	B	D	A	D	D	C
Que.	21	22	23	24	25	26	27	28	29	
Ans.	D	C	B	A	D	C	D	C	D	

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	C,D	A,B,C	B,C	A,B	A,B,C,D	A,B,C	B,C	A,D	D
Que.	11	12	13	14	15					
Ans.	D	B	B	A,B,C	(A)-R ; (B)-Q,S ; (C)-S ; (D)-P					

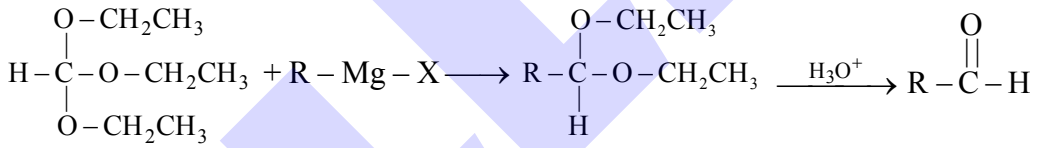
EXERCISE # S-I

1. **(A) P,Q,S ; (B) P,Q,S ; (C) P,Q,S ; (D) P,Q,S**

2. **3 > 2 > 1 > 4**

3. (C) **4.** (A)

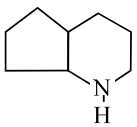
5.



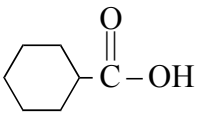
Ethyl orthoformate Acetal aldehyde

6. H^+/Br_2 ; H_2 / Ni ; NaOH

7. (a) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-NH-CH_2-CH_3$ (b)



(c) $CHCl_3$ +



8. (B,C) **9.** (B)

10. (a) (i) $KMnO_4$, (ii) $\underset{\displaystyle CH_2-OH}{\underset{|}{CH_2-OH}}$, (iii) $LiAlH_4$, (iv) $H_3O^{\oplus}$

(b) (i) $\underset{\displaystyle CH_2-OH}{\underset{|}{CH_2-OH}}$, (ii) $NaBH_4$, (iii) H_3O^+, (c) (i) $\underset{\displaystyle CH_2-OH}{\underset{|}{CH_2-OH}}$, (ii) $Ph_3P{=}CH{-}Et$

(d) (i) H_2, Ni (e) NH_2NH_2/H_2O_2 (f) $NaBH_4$

11. **(B)** **12.** **(D)**

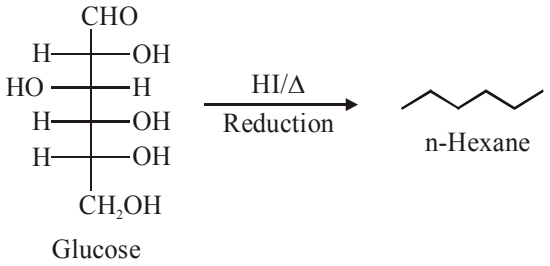
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hlde-Alc & Et - Hydr - Carb - Oxi - Red - Ald & Ket (OC)\Eng\04_Carbonyl\01_Theory+Ex.p65"

EXERCISE # MAINS

1.	**(4)**	**2.**	**(2)**	**3.**	**(1)**	**4.**	**(2)**	**5.**	**(4)**
6.	**(2)**	**7.**	**(4)**	**8.**	**(3)**	**9.**	**(4)**	**10.**	**(2)**
11.	**(4)**	**12.**	**(4)**	**13.**	**(4)**	**14.**	**(3)**	**15.**	**(1, 3)**
16.	**(2)**	**17.**	**(1)**	**18.**	**(1)**	**19.**	**(2)**	**20.**	**(1)**
21.	**(4)**	**22.**	**(1)**	**23.**	**(3)**				

24. (4)

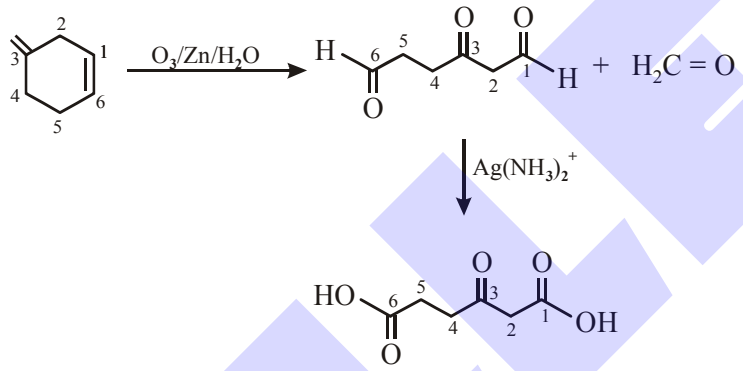
Sol.



25.	**(1)**	**26.**	**(2)**	**27.**	**(1)**	**28.**	**(2)**

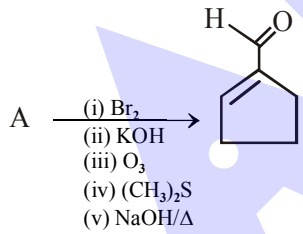
29. (1)

Sol.



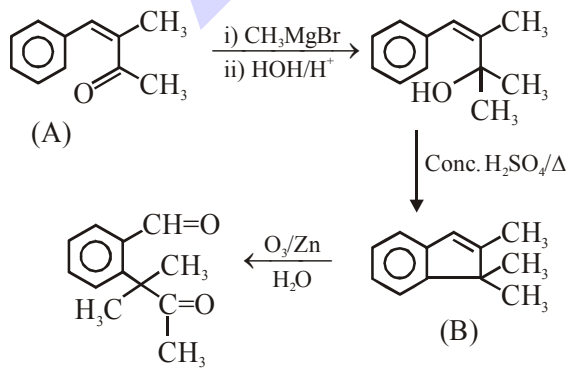
30. (3)

Sol.



31. (4)

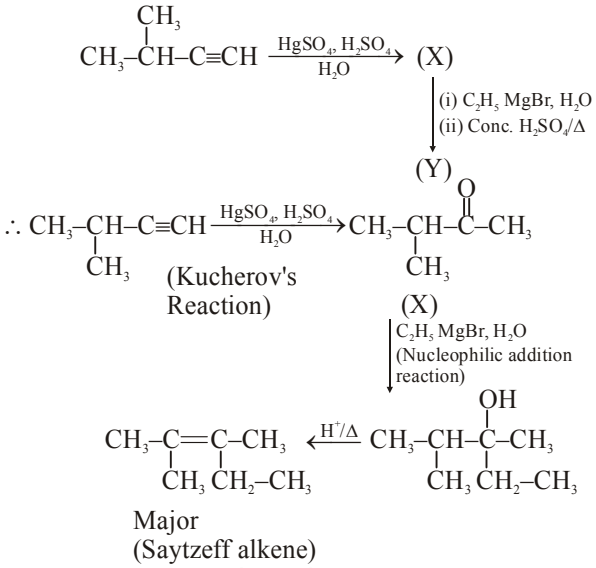
Sol.



node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Hdb-Alc.&Et.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket.(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65"

32. (3)

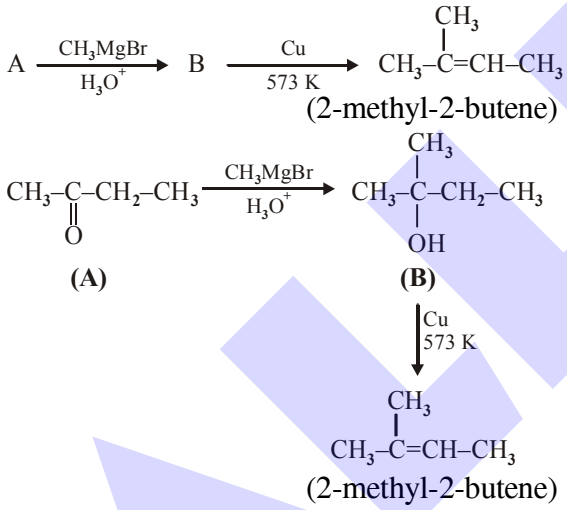
Sol.



33.	**(3)**	**34.**	**(4)**	**35.**	**(2)**	**36.**	**(3)**	**37.**	**(4)**
38.	**(1)**	**39.**	**(1)**	**40.**	**(3)**	**41.**	**(1)**	**42.**	**(4)**
43.	**(2)**	**44.**	**(3)**	**45.**	**(4)**	**46.**	**(1)**	**47.**	**(2)**
48.	**(3)**	**49.**	**(2)**						

50. (66.65 to 66.70)

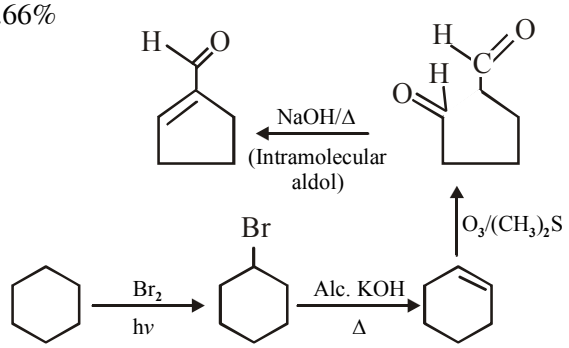
Sol.



C ⇒ 12 × 4 = 48
H ⇒ 8 × 1 = 8
O ⇒ 16 × 1 = 16

Total 72

$$\% \text{ of C} = \frac{48}{72} \times 100 = 66.66\%$$



51. (2) **52. (3)** **53. (3)** **54. (3)**

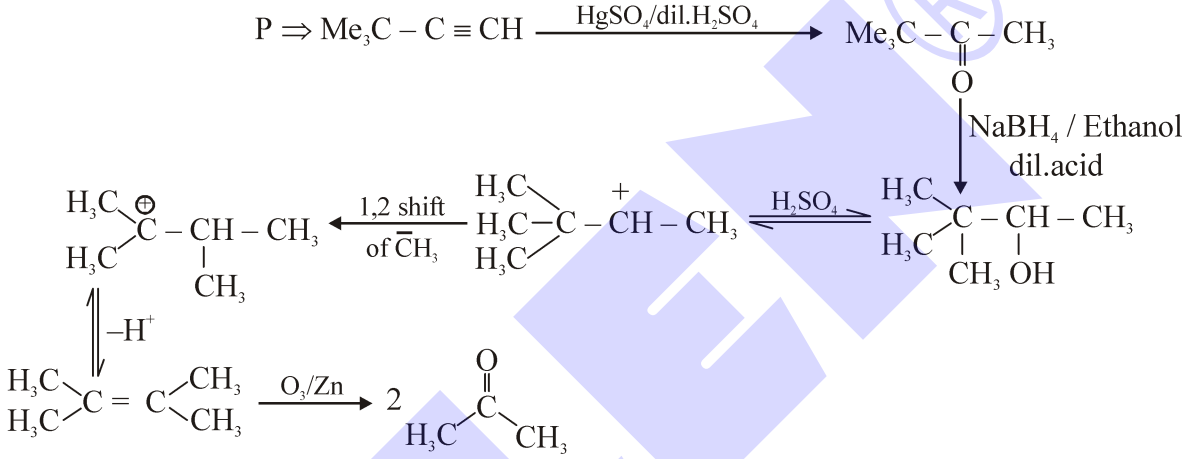
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdlo-Alc & Et - Hydr.-Carb.-Oxi.-Red.- Ald & Ket-(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65

EXERCISE # IIT-QUESTIONS

1.	**(B)**	**2.**	**(A)**	**3.**	**(C)**	**4.**	**(A)**
5.	**(C)**	**6.**	**(D)**	**7.**	**(C)**	**8.**	**(C)**
9.	**(B)**	**10.**	**(A)**	**11.**	**(C)**		
12.	**(A) - P,S ; (B) - Q ; (C) - Q,R,S ; (D) - Q,R**			**13.**	**(B)**	**14.**	**(A)**
15.	**(D)**	**16.**	**(B)**	**17.**	**(A)**	**18.**	**(B)**

19. **(D)**

20. **(B)**



21. **(C)**

Sol. $CH_3CH=O + \bar{O}H \rightleftharpoons \bar{C}H_2 - CH=O \xrightleftharpoons{H-C(=O)-H} H_2C-CH(CH_2OH)-CH=O$ (1 time aldol)

$\xrightleftharpoons{\bar{O}H}$ $\ominus CH_2(CH_2OH)-CH=O \xrightleftharpoons{H-C(=O)-H}$ $HOH_2C-CH(CH_2OH)-CH=O$ (2 time aldol)

$\xrightleftharpoons{\bar{O}H}$ $HOH_2C-\ominus C(CH_2OH)_2-CH=O \xrightleftharpoons{H-C(=O)-H}$ $HOCH_2-C(CH_2OH)_2-CH=O$ (3 times aldol)

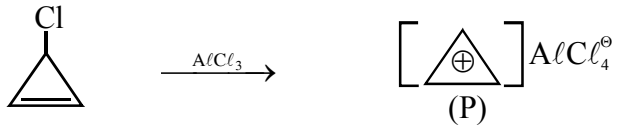
$H-C(=O)-H + \ominus OH \rightleftharpoons HOH_2C-C(CH_2OH)_2-CH_2OH + H-C(=O)-O^{\ominus}$

(crossed cannizaro product)

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.&Et-Hydr.-Carb-Oxi.-Red.-Ald&Ket-(OC)\Eng\04_Carbonyl\101_Theory+Ex.p65

22. (A,B,C,D)

Sol.



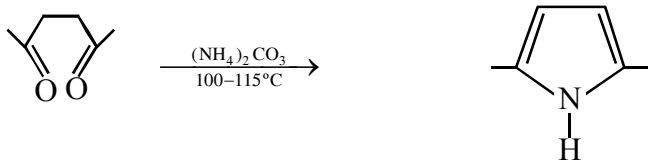
(P)

2π electron (delocalised)
(Aromatic Nature)



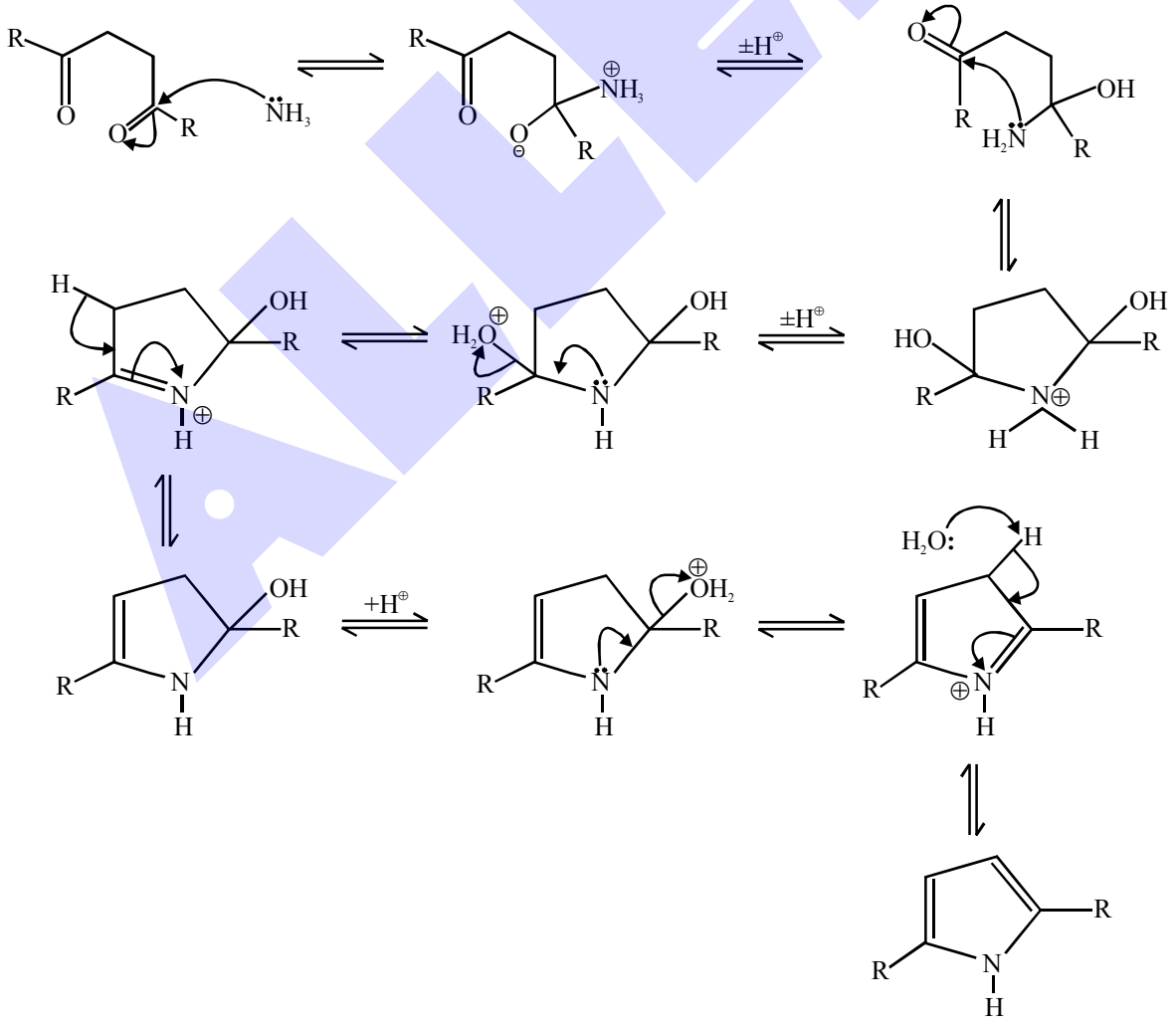
(Q)

6π electron (delocalised)
(Aromatic Nature)



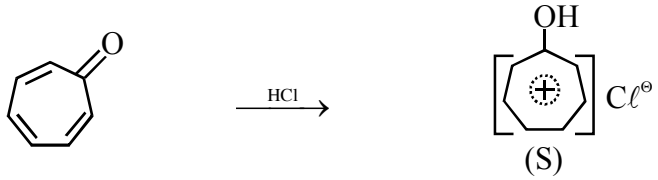
(R) 6π-electron (delocalised)
(Aromatic nature)

Mechanism :



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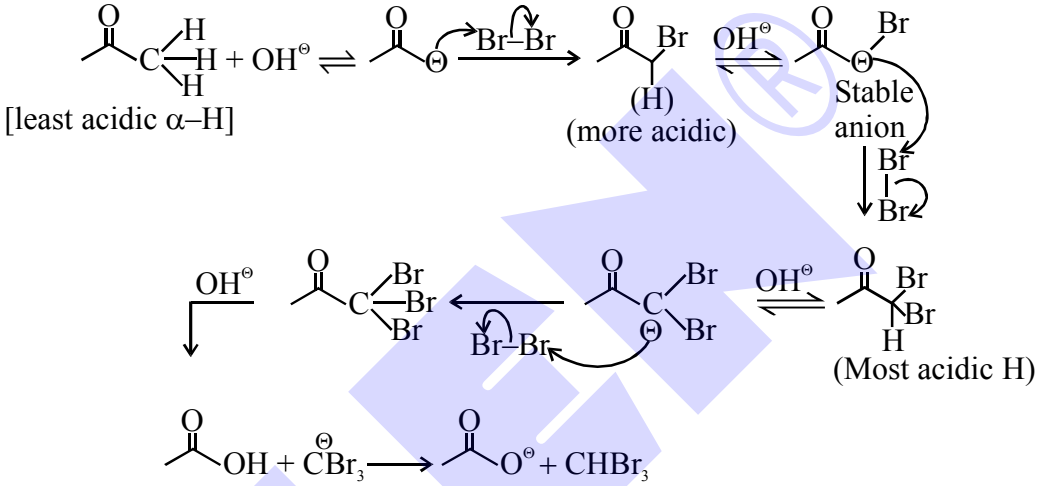
(S)

6π electron (delocalised)
(Aromatic Nature)

23. (C)

Sol. Reaction I : CH_3COCH_3 (1.0 mol) $\xrightarrow[\text{aqueous/जलीय NaOH}]{Br_2\,(1.0\text{ mol})}$ $CH_3COO^{\ominus}Na^{+} + CHBr_3$

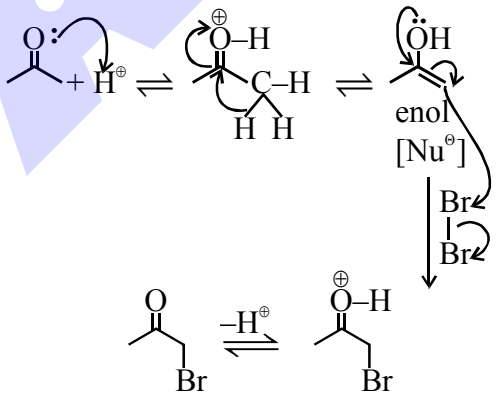
Mechanism



In basic medium halogenation dose not stop with replacement of just one hydrogen and poly halogenation takes place because α-haloketones are more reactive towards base and haloform reaction takes place In above reaction Br_2 is limiting agents.

Reaction II : $CH_3COCH_3 \xrightarrow[MeCOOH]{Br_2(1\text{ mole})} CH_3COCH_2Br$ Monobromination

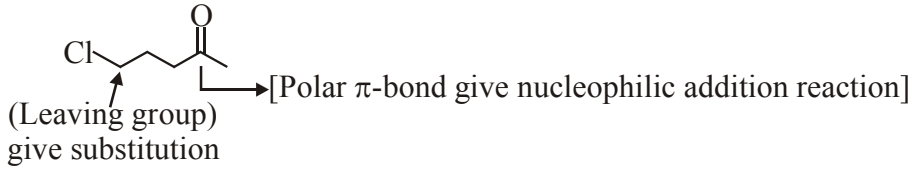
Mechanism : $CH_3COCH_3 \xrightarrow[MeCOOH]{Br_2(1\text{ mole})} CH_3COCH_2Br$ Monobromination



Further bromination is less favourable because of less amount of Br_2

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod#Halo-Alc.&Et.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\04_Carbonyl\101_Theory+Ex.p65

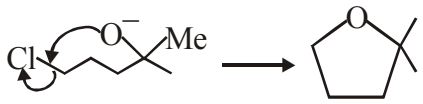
24. (D)



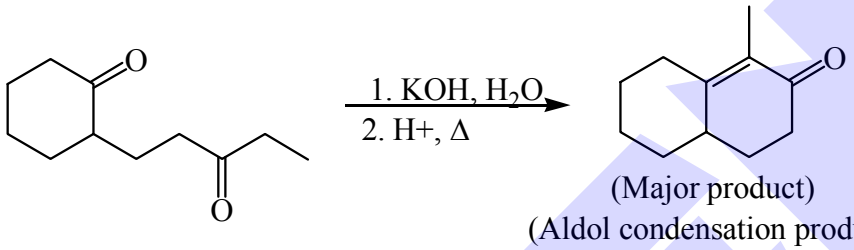
(i) Grignard prefer to give nucleophilic addition on polar π-bond and form anion intermediate.



(ii) In next step anion give intramolecular nucleophilic substitution reaction & form 5 membered ring.

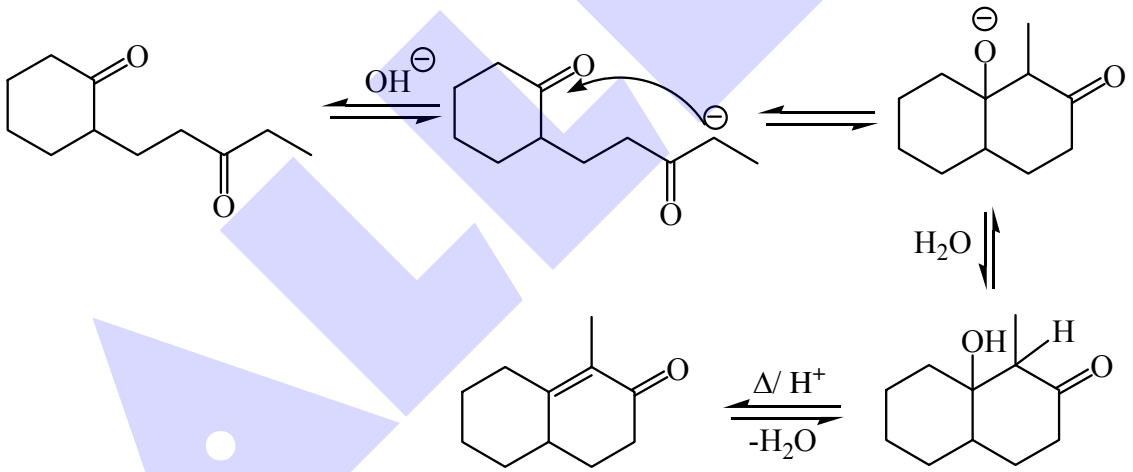


25. **(A)**



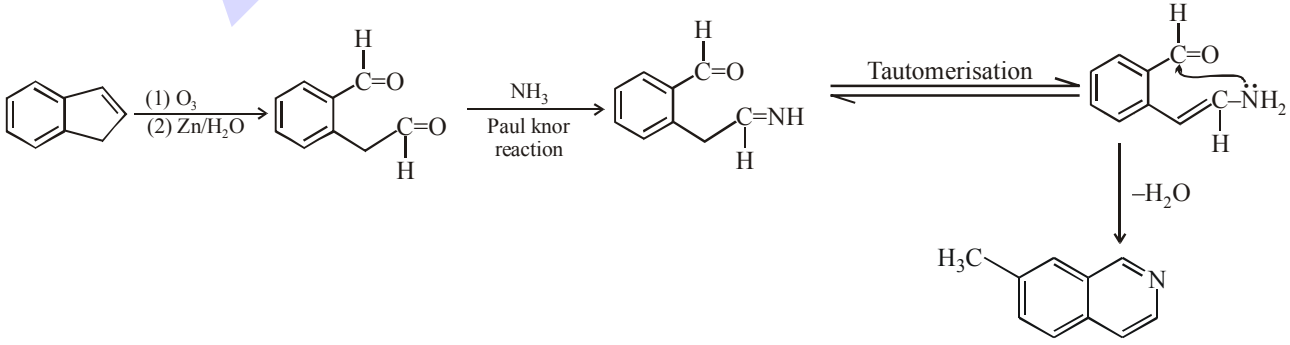
(Major product)
(Aldol condensation product)

Mechanism :



26. **Ans.**(A)

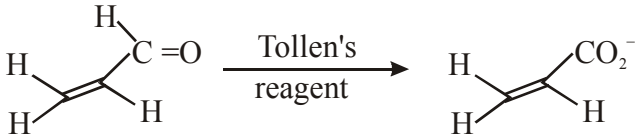
Sol.



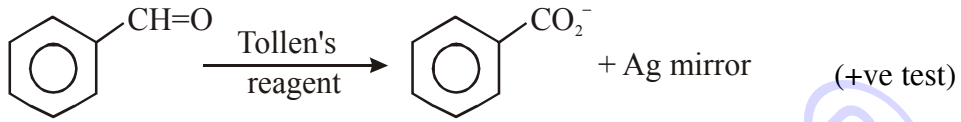
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdb-Alc & Et - Hydr.-Carb.-Oxi.-Red.- Ald & Ket-(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65

27. (A,B,C)

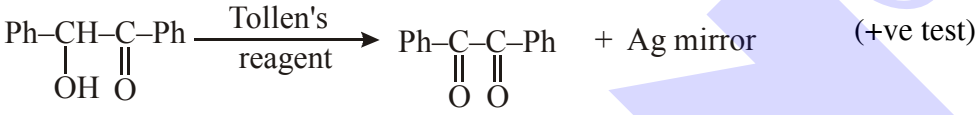
Sol. Tollens's test is given by compounds having aldehyde group. Also α-hydroxy carbonyl gives positive tollen's test.

(A) 

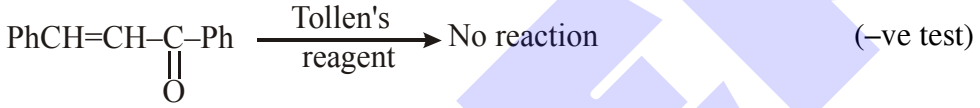
Acraldehyde

(B) 

Benzaldehyde

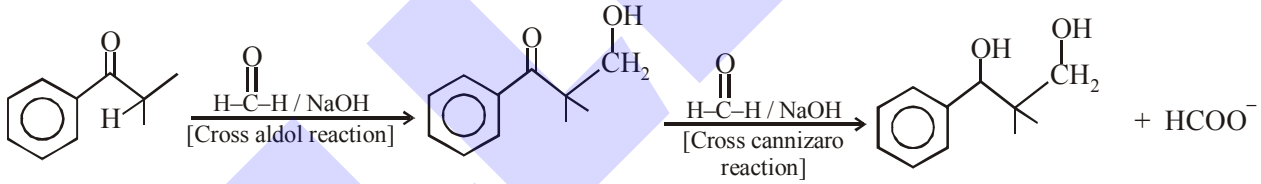
(C) 

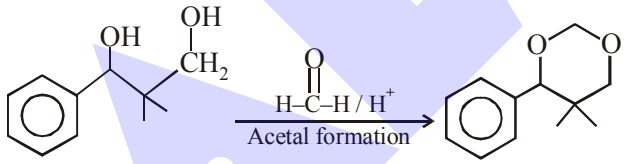
Benzoin

(D) 

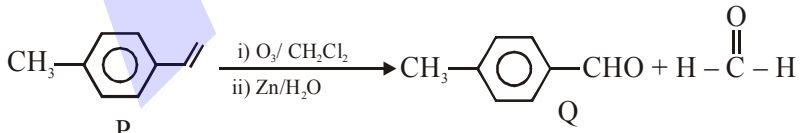
28. (A)

Sol.



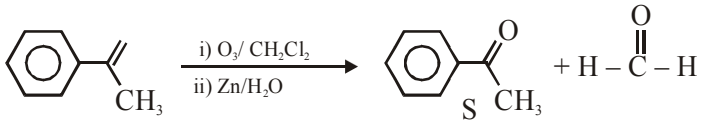


29. (A,C)

Sol. **(A)** 

P Q

give cannizzaro reaction
but no haloform



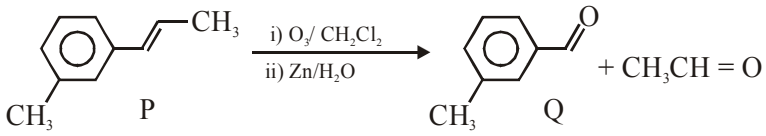
R S

no cannizzaro
but give haloform

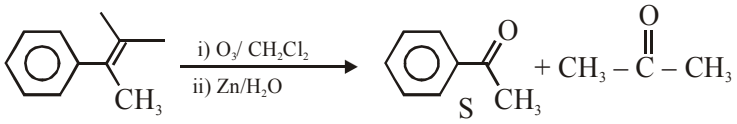
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#\Hdb-Alc.&Et.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket.(OC)\Eng\04_Carbonyl\101_Theory+Ex.p65"

(B) Product of ozonolysis of R is having 9 carbon.

(C)



give cannizzaro reaction
but no haloform

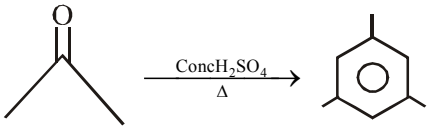


no cannizzaro
but give haloform

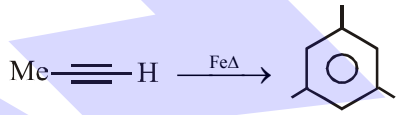
(D) Product of ozonolysis of R is having 9 carbon.

30. **(A,B,D)**

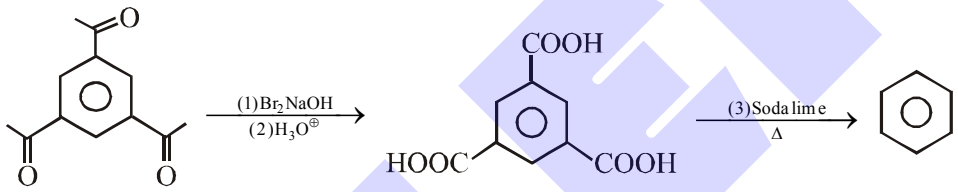
Sol. (A)



(B)



(C)



(D)



31. **(D)**

ALLEN®

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hlde-Alc & Et - Hydr.-Carb.-Oxi.-Red.- Ald & Ket-(OC)\Eng\04_Carbonyl\01_Theory+Ex.p65